Universidad Politécnica de Madrid

Escuela Técnica Superior de Ingenieros Informáticos

European Masters in Human Computer Interaction & Design

Data Mining Report on Credit Card Default

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Data Mining

April 17

A report on using Data Mining to help Companies predict who will default on credit cards

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Machine Learning Algorithms to predict who will default

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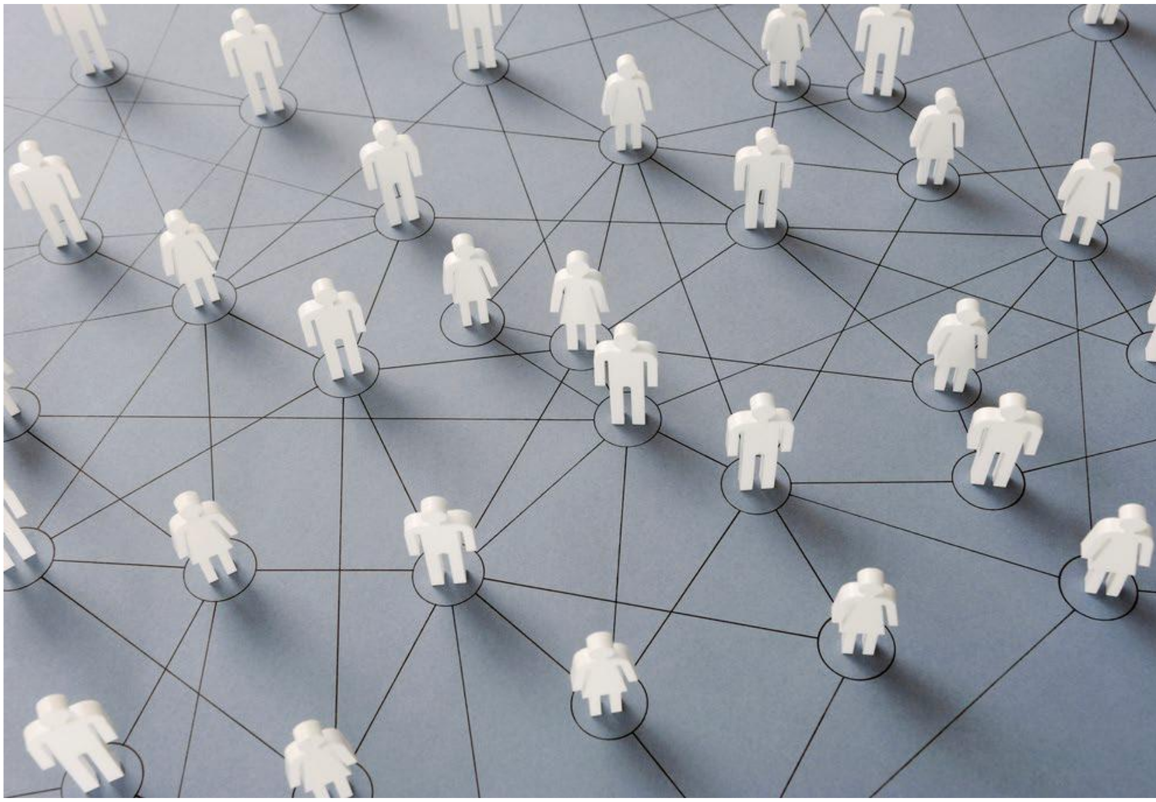
Evaluation

Reduce Default Rate
Segmented Customers
Attract & Retain Customers
All based on Credit Card



Section 1: Business understanding

Business Goals

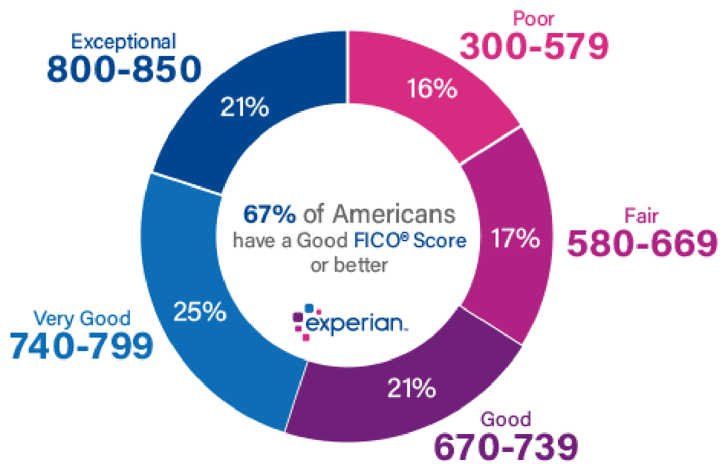
1. **Reduce credit card default rates by identifying customers at high risk of defaulting.**
2. **Increase customer satisfaction and retention by understanding their financial behavior and needs?**
3. **Develop targeted marketing campaigns to attract new customers and retain existing ones based on their credit card usage.**
   Example: After finding out we can know which people are going to default on Credit Card, and which will not. We can run different business campaigns targeting each. For people who are likely to default, we can incentivize them to clear the debit on time and we can give them good offers. For those people, who will not default, we can offer them a higher credit limit and loan opportunity to take up more credit, to increase the business of the company.

Business Questions

1. **What is the proportion of customers who default on their credit card payments?**
2. **What are the demographic characteristics (age, gender, education, marital status) of customers who default on their credit card payments?**
3. **What is the relationship between past payment history and default rates?**
4. **How can we identify customers who are likely to default in the future?**
5. **How can we segment customers based on their financial behavior and credit card usage?**

Why buy our product?

I. Credit Card companies rely on customers credit scores to determine if they are eligible for services like credit limit increases and loan opportunities.



II. Default Rate is a big issue for Credit Card Companies leading to revenue losses and a decline in customer satisfaction. Here are the biggest corporations in the Banking Industry at every level.

III. We are offering our data mining techniques and applying it on University of California Irvine's Dataset of Default on Credit Card to identify customers at high risk of default and develop targeted marketing campaign to attract and retain customers based on their credit card use. There are multiple benefits of working with the dataset by UCI

   a. The Dataset has more than double digit citations on google scholar
   b. Contains Large amount of real-world credit card data
   c. Data has been pre-processed and is ready to use
   d. Data has rich features like demographic info and past payment history
   e. Publicly Available Dataset, allowing our research to be transparent and verifiable

IV. Finally, to summarize there are multiple reasons to buy our product:

   a. Improve Decision Making
   b. Increase Marketing Campaigns
   c. Revenue Loss prevention
   d. Risk Identification
   e. Competitive Edge
   f. Help meet Regulatory Requirements

V. We really recommend reading the Overview, Data Insights, Customer Segment and Marketing Campaign, Predictive Analysis and Evaluation with Call to Action to learn about the most important sections of the report.

Data Set Description

The UCI Default of Credit Card Clients dataset contains information on credit card clients in Taiwan from April to September 2005. The datasSet includes 30,000 records and 25 variables, including demographic variables, credit card usage and payment history, and binary variable indicating whether the client defaulted on their payment in the following month.

Descriptive Analysis for Dataset (Questions about what happened):

- When: The dataset records the credit card usage and payment history of clients in Taiwan from April to September 2005.
- How many: The dataset contains 30,000 records of credit card clients.
- How often: The dataset records the credit card usage and payment history of clients monthly from April to September 2005.
- Where: The dataset was collected from credit card clients in Taiwan.

Data Set Description (As mentioned on website)

This research employed a binary variable, default payment (Yes = 1, No = 0), as the response variable. This study reviewed the literature and used the following 23 variables as explanatory variables:

X1: Amount of the given credit (NT dollar): it includes both the individual consumer credit and his/her family (supplementary) credit.

X2: Gender (1 = male; 2 = female).

X3: Education (1 = graduate school; 2 = university; 3 = high school; 4 = others).

X4: Marital status (1 = married; 2 = single; 3 = others).

X5: Age (year).

X6 - X11: History of past payment. We tracked the past monthly payment records (from April to September, 2005) as follows: X6 = the repayment status in September, 2005; X7 = the repayment status in August, 2005; . . .;X11 = the repayment status in April, 2005. The measurement scale for the repayment status is: -1 = pay duly; 1 = payment delay for one month; 2 = payment delay for two months; . . .; 8 = payment delay for eight months; 9 = payment delay for nine months and above.

X12-X17: Amount of bill statement (NT dollar). X12 = amount of bill statement in September, 2005; X13 = amount of bill statement in August, 2005; . . .; X17 = amount of bill statement in April, 2005.

X18-X23: Amount of previous payment (NT dollar). X18 = amount paid in September, 2005; X19 = amount paid in August, 2005; . . .;X23 = amount paid in April, 2005.

We remove the first row, X1-X23 numbers that represent the variables, and use the common names like Amount of given credit, gender, education, martial status, Age, History of Past Payment.

Links:
https://archive.ics.uci.edu/ml/datasets/default+of+credit+card+clients

https://www.kaggle.com/datasets/uciml/default-of-credit-card-clients-dataset

Overview

Data Mining Goal: Use predictive modeling techniques to identify customers who are at high risk of defaulting on their credit card payments and develop targeted strategies to reduce default rates.

Descriptive: Counts, Averages, Percentages, Variations, and other statistical tools will help in understanding the distribution of the data, identifying trends and patterns, and summarizing the key characteristics of the dataset.

- **What are the demographic characteristics of customers more likely to default on payments?**
- **What is the payment history of customers who defaulted on their payments?**
- **Are there any patterns or trends in the credit card use behavior who defaulted?**
- **What can we do to prevent customers from defaulting, such as new payment plans or offering credit advice?**

Diagnostic: Diagnostic analysis will help in identifying the reasons behind credit card defaults by analyzing the relationships between the explanatory variables and the response variable.

Predictive: Predictive modeling techniques such as logistic regression, decision trees, and random forest will help in predicting the likelihood of defaulting for new customers based on their demographic characteristics and past payment history.

Prescriptive: Prescriptive analysis will provide recommendations for actions to take in order to reduce credit card defaults. For example, offering financial education or debt management programs to customers at high risk of defaulting.

Segmentation and Clustering: Segmenting customers based on their financial behavior and credit card usage can help in identifying groups of customers with similar characteristics and needs and developing targeted strategies to reduce default rates.

RFM Segmentation and Analysis: RFM analysis can help in identifying the most valuable customers based on their recency, frequency, and monetary value, and developing targeted marketing campaigns to retain them. However, this may not be applicable in the context of credit card default analysis, as the focus is on identifying customers who are at high risk of defaulting rather than the most valuable customers. Our dataset doesn't include transactions, so **RFM can't be applied directly.** We can use the attributes of dataset to identify the characteristics of customers who are more likely to default on credit card payments.

If we talk about RFM Analysis, it is done for identifying and analyzing different customer groups on their transactions. It includes recency, frequency and monetary value of purchase to segment into groups.

Business Goal: Reduce Default Rate of Credit Card by identifying customers at high risk of defaulting

Can we predict which customers will default on credit cards and how? based on previous context

Yes, based on the previous context and the variables provided, we can build a **predictive model** to identify which customers are likely to default on their credit cards. The response variable in the dataset is a binary variable indicating whether a customer defaulted on their credit card payment or not, which makes it suitable for building a classification model.

By exploring the relationships between the explanatory variables and the response variable, we can identify the factors that are most strongly associated with credit card default and use this information to build a predictive model. Some of the factors that may be relevant for predicting credit card default include the amount of credit given, the customer's gender, education, marital status, and their past payment history.

In terms of methodology, we can use machine learning algorithms such as **logistic regression, decision trees, or random forests** to build a predictive model. These algorithms can be **trained on a subset of the dataset and** then **evaluated on a separate test** set to assess their predictive performance.

Once a model with satisfactory predictive performance is developed, it can be used to identify which customers are most likely to default on their credit card payments, which can be used to inform business decisions such as targeted marketing campaigns or credit risk management strategies.

Section 2: Data Understanding: Descriptive and Diagnostic Analysis

Data Preparation

- Create a single dataset
- Check for NaN/Null Values
- Check the data for outliers

As we see, there are no NaN values in the dataset.

When we do the statistics node and see the aggregate of Education for all the units, we find:

Education =0, 14 units

Education	Count
0	14
1	10585
2	14030
3	4917
4	123
5	280
6	51

Complete fields (%): 100% Complete records (%): 100%

Field	Measurement	Outliers	Extremes	Action	Impute Missing	Method	% Complete	Valid Records	Null Value	Empty String	White Space	Blank Value
ID	Continuous	0	0	None	Never	Fixed	100	30000	0	0	0	0
LIMIT_BAL	Continuous	129	1	None	Never	Fixed	100	30000	0	0	0	0
SEX	Flag	--	--	--	Never	Fixed	100	30000	0	0	0	0
EDUCATION	Nominal	--	--	--	Never	Fixed	100	30000	0	0	0	0
MARRIAGE	Flag	--	--	--	Never	Fixed	100	30000	0	0	0	0
AGE	Continuous	141	0	None	Never	Fixed	100	30000	0	0	0	0
PAY_0	Continuous	102	39	None	Never	Fixed	100	30000	0	0	0	0
PAY_2	Continuous	124	33	None	Never	Fixed	100	30000	0	0	0	0
PAY_3	Continuous	97	53	None	Never	Fixed	100	30000	0	0	0	0
PAY_4	Continuous	104	65	None	Never	Fixed	100	30000	0	0	0	0
PAY_5	Continuous	101	63	None	Never	Fixed	100	30000	0	0	0	0
PAY_6	Continuous	62	67	None	Never	Fixed	100	30000	0	0	0	0
BILL_AMT1	Continuous	544	142	None	Never	Fixed	100	30000	0	0	0	0
BILL_AMT2	Continuous	528	142	None	Never	Fixed	100	30000	0	0	0	0
BILL_AMT3	Continuous	524	137	None	Never	Fixed	100	30000	0	0	0	0
BILL_AMT4	Continuous	532	148	None	Never	Fixed	100	30000	0	0	0	0
BILL_AMT5	Continuous	506	145	None	Never	Fixed	100	30000	0	0	0	0
BILL_AMT6	Continuous	508	143	None	Never	Fixed	100	30000	0	0	0	0
PAY_AMT1	Continuous	195	195	None	Never	Fixed	100	30000	0	0	0	0
PAY_AMT2	Continuous	164	143	None	Never	Fixed	100	30000	0	0	0	0
PAY_AMT3	Continuous	159	203	None	Never	Fixed	100	30000	0	0	0	0
PAY_AMT4	Continuous	188	208	None	Never	Fixed	100	30000	0	0	0	0
PAY_AMT5	Continuous	187	227	None	Never	Fixed	100	30000	0	0	0	0
PAY_AMT6	Continuous	186	253	None	Never	Fixed	100	30000	0	0	0	0
default paym...	Flag	--	--	--	Never	Fixed	100	30000	0	0	0	0

Figure: Shows Data Audit verifying the records are valid with no empty strings, white space, and other checks.

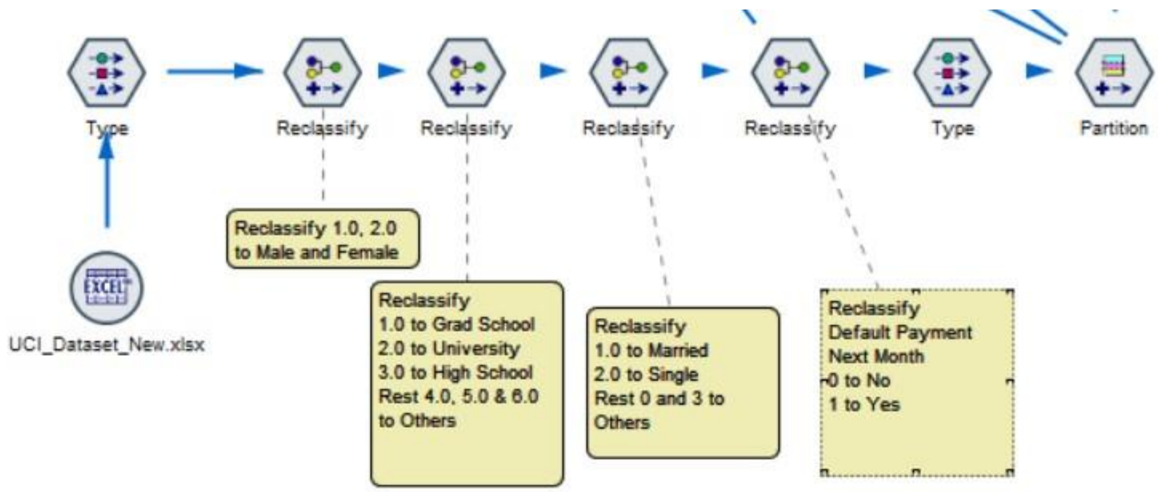


Figure: Here, we explain how we Reclassify the initial Dataset, so it is easier to do operations

Here, we can see that data values 4,5,6 and 0 correspond to Others, and not really statistically significant. We decided to keep all the values of the dataset, as we believe it's important to be inclusive regardless of someone's education or other demographic data, as their credit history and other attributes are just as important.

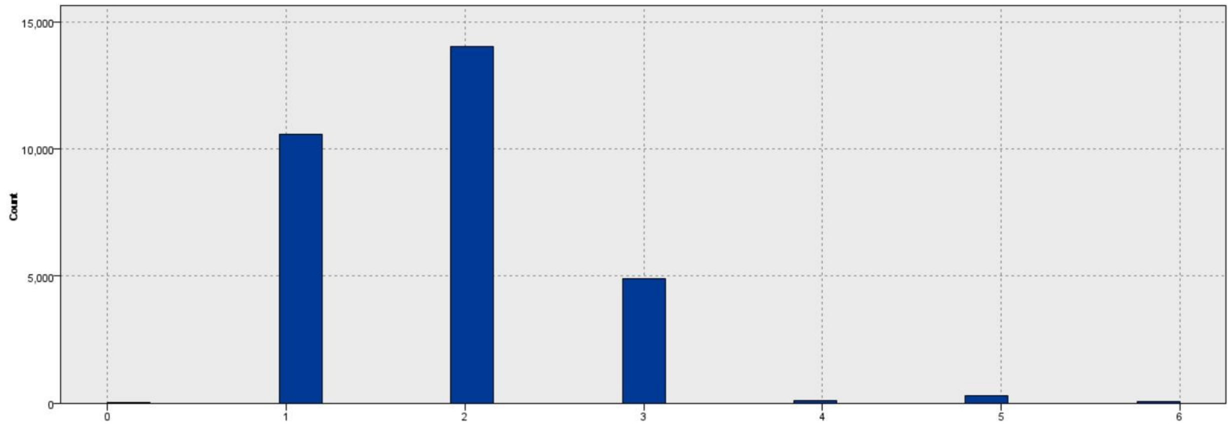


Figure: From L to R, 0 to 6. Units 0, 4, 5 and 6 are reclassified to OTHERS.

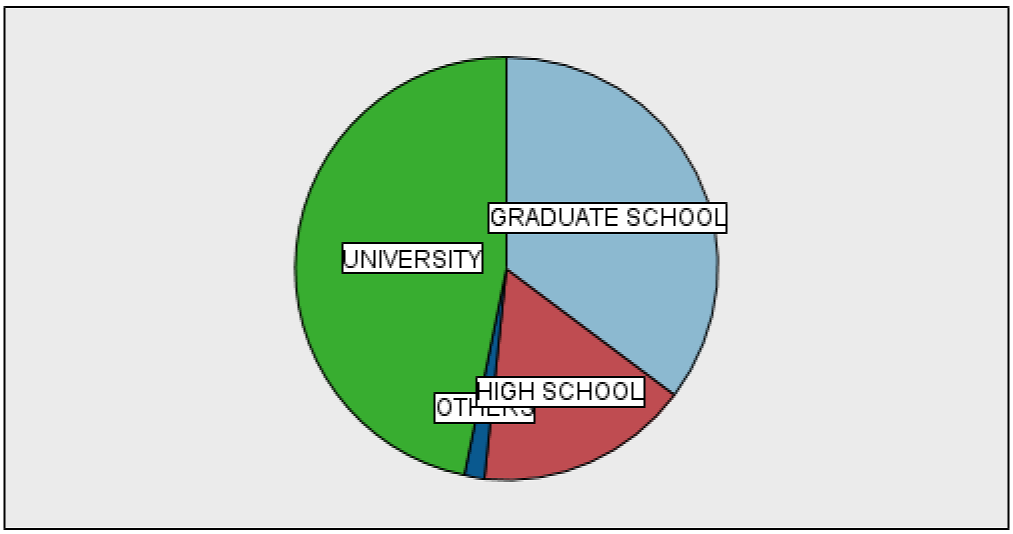


Figure: Chart after reclassification. Majority is University, then Graduate school and then High school.

Similarly, we find values 0, 1 and 2 in Marriage. Although, 0 values are very few and can be removed, we believe that the overall value of that Individual might be insightful so we keep all the values here too, and don't delete any rows.

Visualizations that can be used in further sections:

I. Scatter Plot
II. Confusion matrix

Skipped:

I. Heat Map
II. ROC Curve
III. Tree

View Data from Analysis Node:

28	$CC-default payment next month		Continuous	0.531	0.862	24,782.92	0.331	0.826	0.082	0.007	0	-2.2
29	$R-default payment next month		Nonimal	--	--	--	--	--	--	--	--	--
30	$RC-default payment next month		Continuous	0.531	0.951	24,706.853	0.42	0.824	0.11	0.012	0.001	-0.9
31	$R1-default payment next month		Nonimal	--	--	--	--	--	--	--	--	--
32	$RC1-default payment next month		Continuous	0.694	0.84	24,735.823	0.146	0.825	0.044	0.002	0	-2.5
33	$R2-default payment next month		Nonimal	--	--	--	--	--	--	--	--	--
34	$RC2-default payment next month		Continuous	0.684	0.84	24,706.633	0.156	0.824	0.048	0.002	0	-2.5
35	$S-default payment next month		Nonimal	--	--	--	--	--	--	--	--	--
36	$SP-default payment next month		Continuous	0.5	1	24,419.652	0.5	0.814	0.085	0.007	0	-2.0
37	$KNN-default payment next month		Nonimal	--	--	--	--	--	--	--	--	--
38	$KNNP-default payment next month		Continuous	0.5	0.833	21,878.833	0.333	0.729	0.118	0.014	0.001	-0.6
39	$N-default payment next month		Nonimal	--	--	--	--	--	--	--	--	--
40	$NC-default payment next month		Continuous	0.5	0.986	24,720.416	0.486	0.824	0.108	0.012	0.001	-1.1

12	PAY_6		Continuous	-2	8	-8,733	10	-0.291	1.15	1.322	0.007	0.94
13	BILL_AMT1		Continuous	-165,580	964,511	1,536,699,927	1,130,091	51,223.331	73,635.861	5,422,239,962.699	425.137	2.66
14	BILL_AMT2		Continuous	-69,777	983,931	1,475,372,255	1,053,708	49,179.075	71,173.769	5,065,705,362.709	410.922	2.70
15	BILL_AMT3		Continuous	-157,264	1,664,089	1,410,394,644	1,821,353	47,013.155	69,349.387	4,809,337,536.505	400.389	3.08
16	BILL_AMT4		Continuous	-170,000	891,586	1,297,888,469	1,061,586	43,262.949	64,332.856	4,138,716,378.347	371.426	2.82
17	BILL_AMT5		Continuous	-81,334	927,171	1,209,342,029	1,008,505	40,311.401	60,797.156	3,696,294,149.754	351.013	2.87
18	BILL_AMT6		Continuous	-339,603	961,664	1,166,152,812	1,301,267	38,871.76	59,554.108	3,546,691,724.498	343.836	2.84
19	PAY_AMT1		Continuous	0	873,552	169,907,415	873,552	5,663.581	16,563.28	274,342,256.086	95.628	14.6
20	PAY_AMT2		Continuous	0	1,684,259	177,634,905	1,684,259	5,921.163	23,040.87	530,881,708.884	133.027	30.4
21	PAY_AMT3		Continuous	0	896,040	156,770,445	896,040	5,225.681	17,606.961	310,005,092.199	101.654	17.2
22	PAY_AMT4		Continuous	0	621,000	144,782,306	621,000	4,826.077	15,666.16	245,428,561.126	90.449	12.9
23	PAY_AMT5		Continuous	0	426,529	143,981,629	426,529	4,799.388	15,278.306	233,426,624.425	88.209	11.12
24	PAY_AMT6		Continuous	0	528,666	156,465,077	528,666	5,215.503	17,777.466	316,038,289.397	102.638	10.6

25	default payment next month		Nonimal	--	--	--	--	--	--	--	--	--
26	Partition		Nonimal	--	--	--	--	--	--	--	--	--
27	$C-default payment next month		Nonimal	--	--	--	--	--	--	--	--	--
28	$CC-default payment next month		Continuous	0.531	0.862	24,782.92	0.331	0.826	0.082	0.007	0	-2.2
29	$R-default payment next month		Nonimal	--	--	--	--	--	--	--	--	--
30	$RC-default payment next month		Continuous	0.531	0.951	24,706.853	0.42	0.824	0.11	0.012	0.001	-0.9
31	$R1-default payment next month		Nonimal	--	--	--	--	--	--	--	--	--
32	$RC1-default payment next month		Continuous	0.694	0.84	24,735.823	0.146	0.825	0.044	0.002	0	-2.5
33	$R2-default payment next month		Nonimal	--	--	--	--	--	--	--	--	--
34	$RC2-default payment next month		Continuous	0.684	0.84	24,706.633	0.156	0.824	0.048	0.002	0	-2.5
35	$S-default payment next month		Nonimal	--	--	--	--	--	--	--	--	--
36	$SP-default payment next month		Continuous	0.5	1	24,419.652	0.5	0.814	0.085	0.007	0	-2.0
37	$KNN-default payment next month		Nonimal	--	--	--	--	--	--	--	--	--

Boxplots:

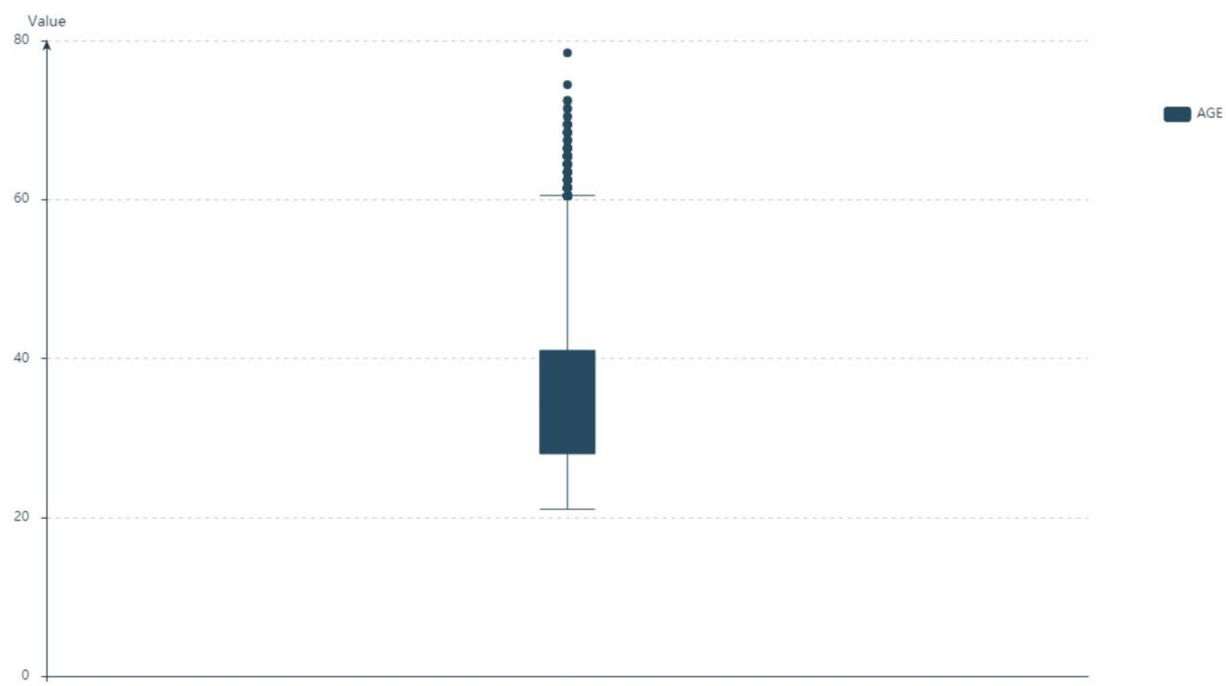


Figure: Boxplot of Age

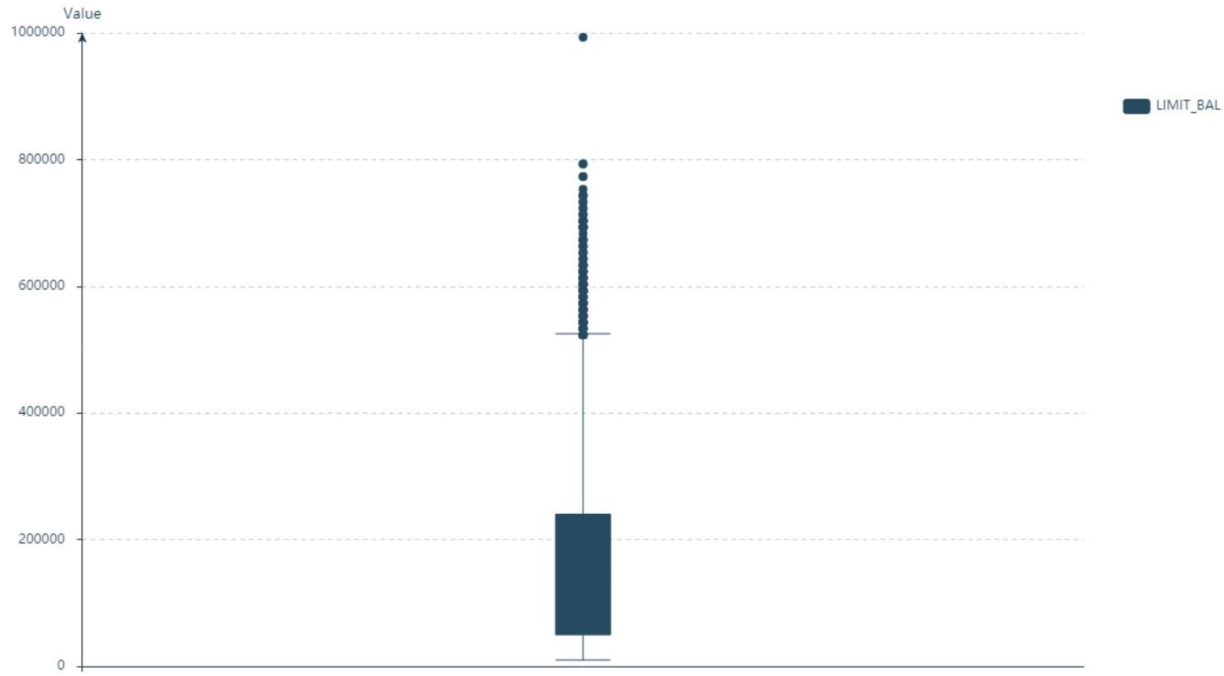


Figure: Boxplot of Limit_Bal

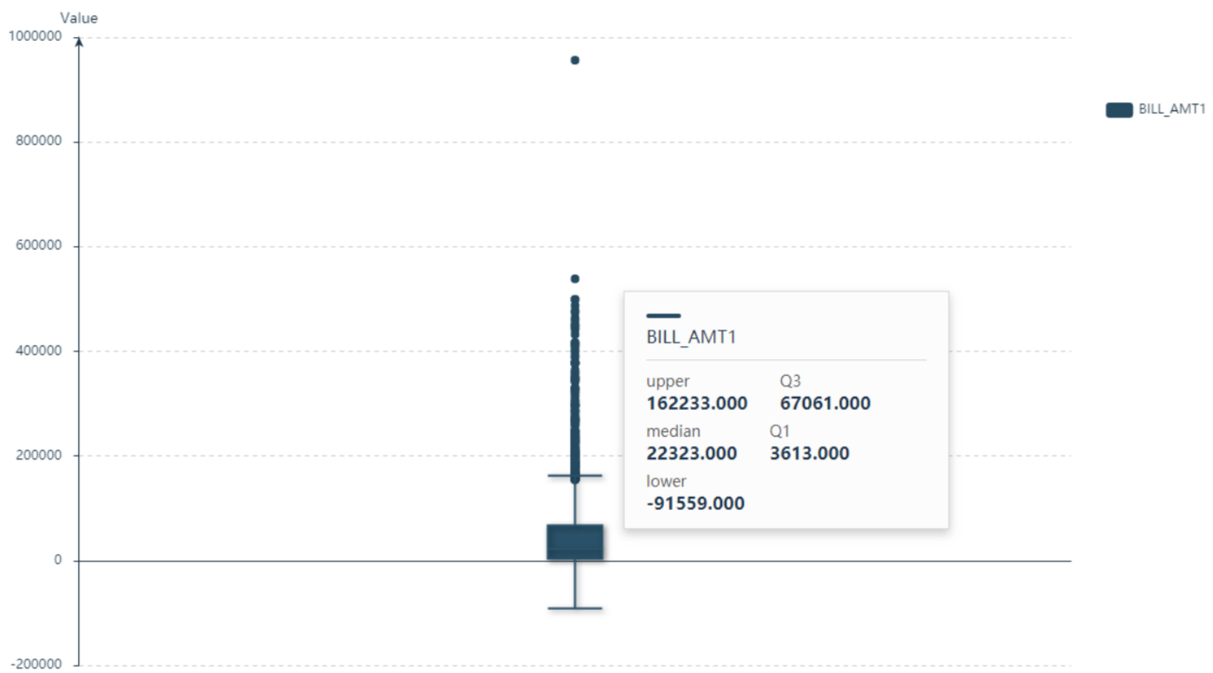


Figure: Boxplot of Bill_Amt1

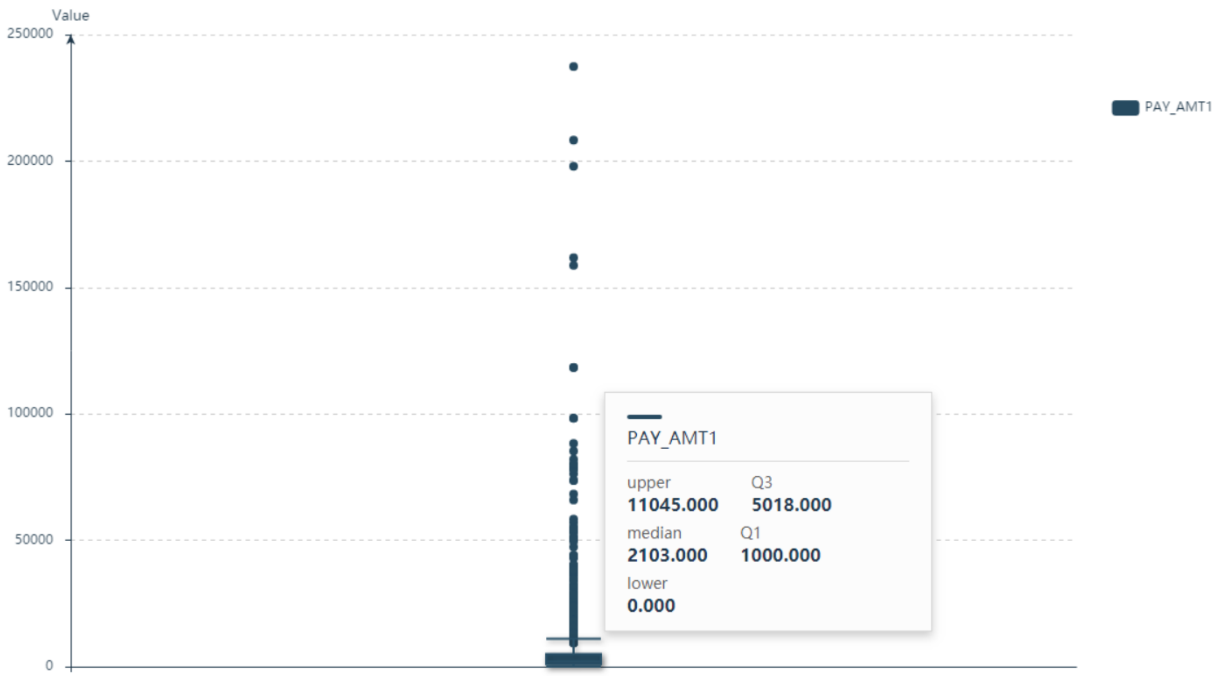


Figure: Boxplot of Pay_Amt1

Outliers:
Outliers are data points that significantly differ from other data points in a dataset. These are data points that are distant from the mean of the dataset and can negatively affect statistical analysis and machine learning models.

We see that our Dataset is large, and the impact of outliers is minimal and their removal will not impact the statistical analysis of our data.

Discussed first in research paper "The Detection of Outliers" by Hawkins (1980)

Data Insight

Customer Data:

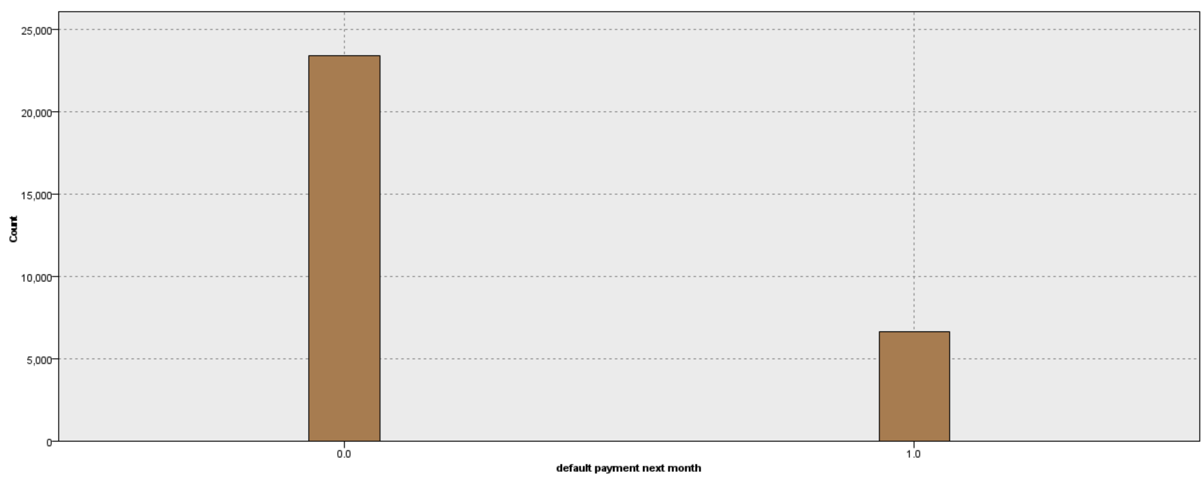
- Credit Amount (New Taiwan Dollar)
- Gender
- Education
- Martial Status
- Age
- Repayment Status/ History of Past Payment

Data	Type	Values
Credit Amount (New Taiwan Dollar)	Continuous	
Sex	Flag	Male, Female
Education	Nominal	High School, University, Other
Marriage	Categorical	Married, Single
Age	Nominal	
Pay	Categorical	
Default	Categorical	1/0

Target?

Default Payment
A binary variable indicating whether a customer has defaulted on the payment of Credit card in the following month (1/0)

Chart on Default Payment:



Target in Business terms:

Which customers are at risk of defaulting on their credit card payments based on their past payment history, demographic information, and credit limit?

Data Insights:
Youngest: 21
Oldest: 79
Average Age : 35

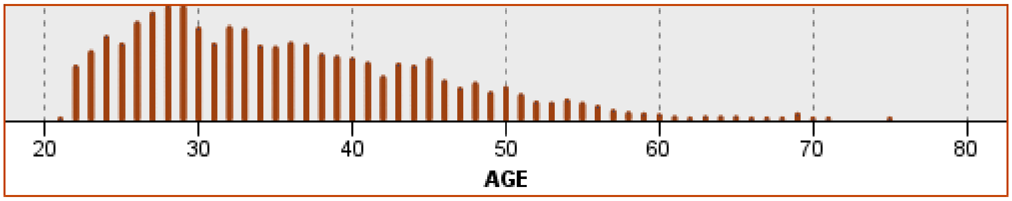


Figure: Provide a dot plot to help visualize the age across the spectrum.

Credit History:

Lowest Credit: 10000.00
Highest credit: 1000000.00
Average Credit: 167484.323

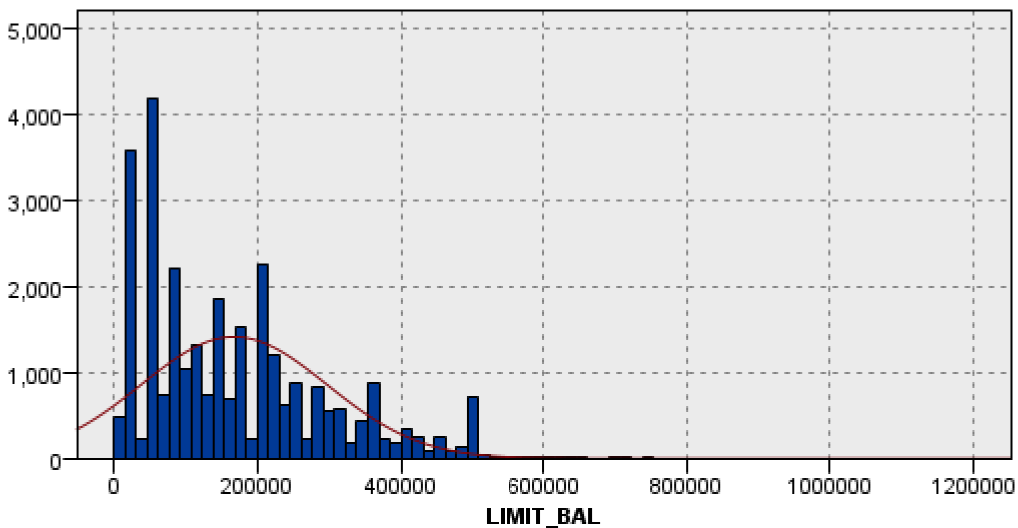


Figure: We use histogram with normal distribution curve to show that it follows a normal distribution although it’s little skewed on one side.

Chart on Education

Education level is mostly graduate school and university, and Other.

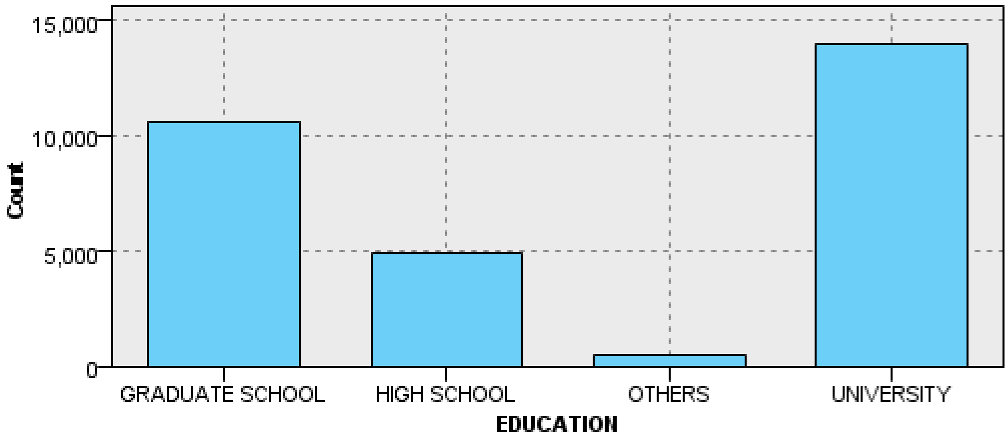
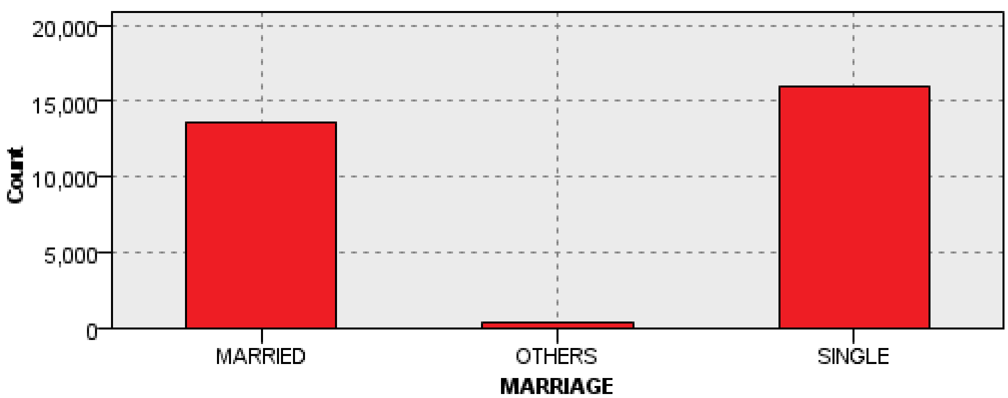


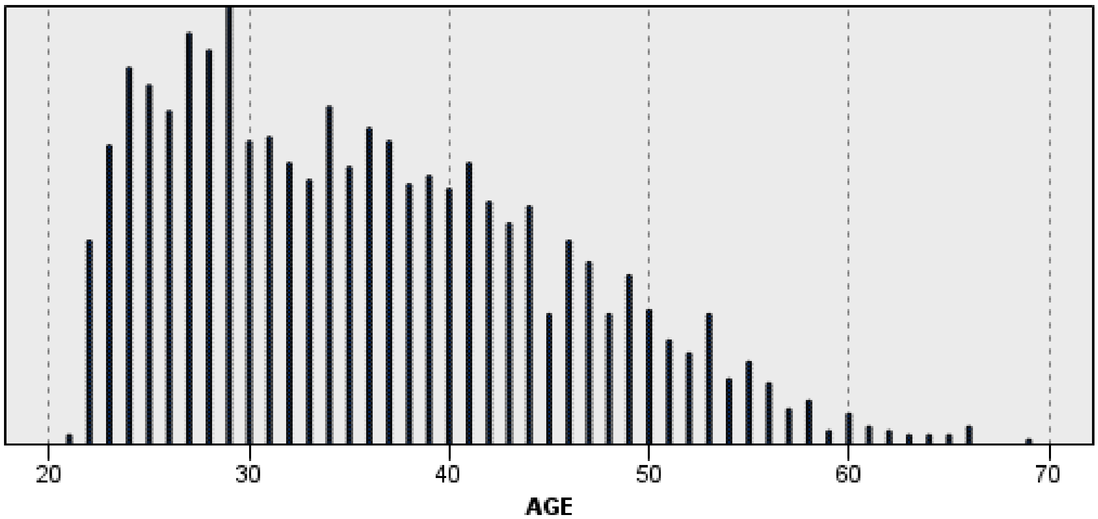
Chart on Martial Status

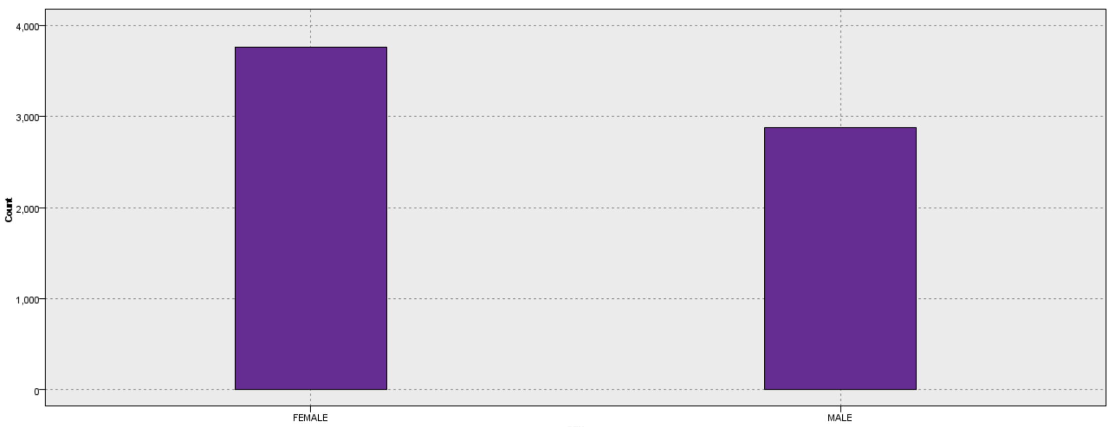


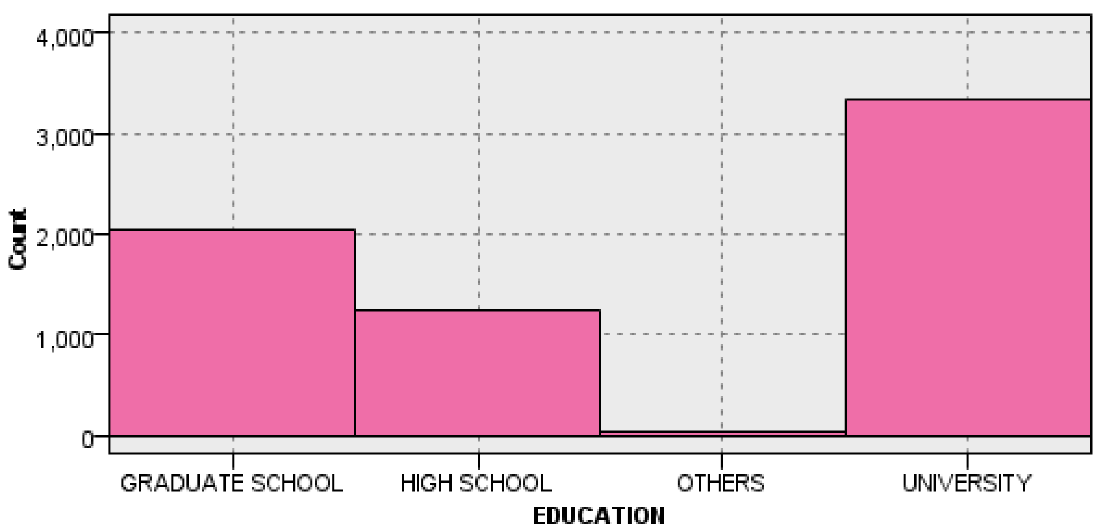
Key Insights from Graph Comparison (Compare Demographic Variables with default next month = 0/1)

- For default next month = 0: The number of customers with a university education (Education = 2) who did not default next month is higher than the number of customers with other education levels.
- For default next month = 1: The number of customers with a high school education (Education = 3) who defaulted next month is higher than the number of customers with other education levels.
- For default next month = 0: The number of females (Sex = 2) who did not default next month is higher than the number of males (Sex = 1) who did not default next month.

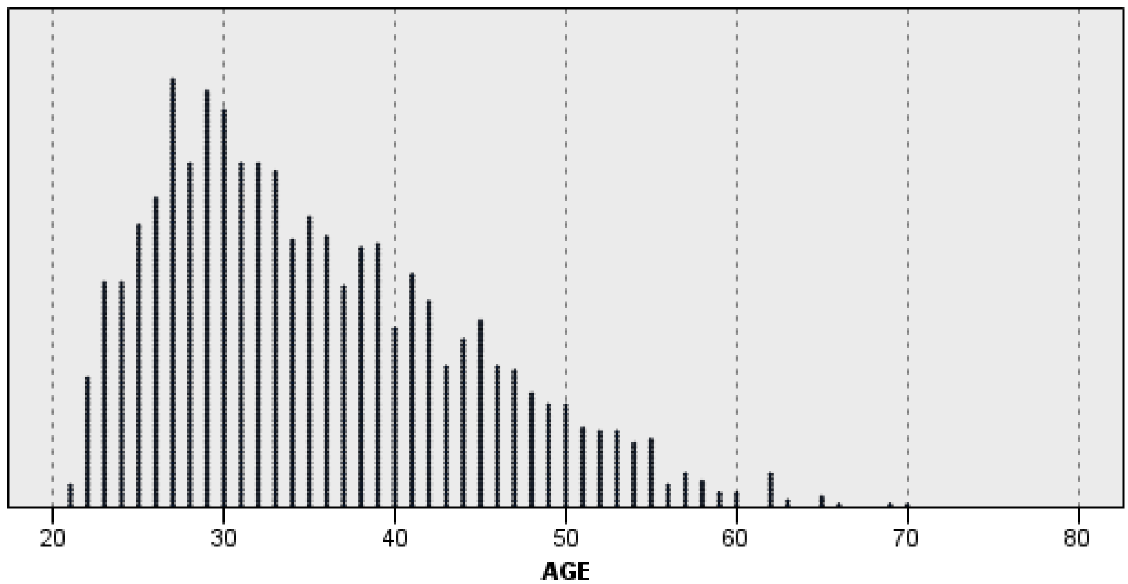
Further, Our analysis For Case Default =1, based on Age, Gender and Education:

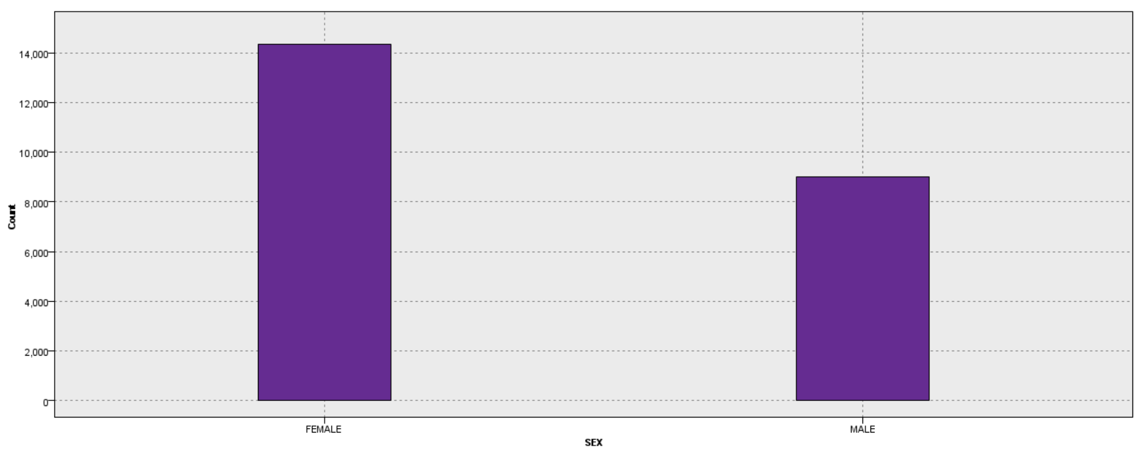


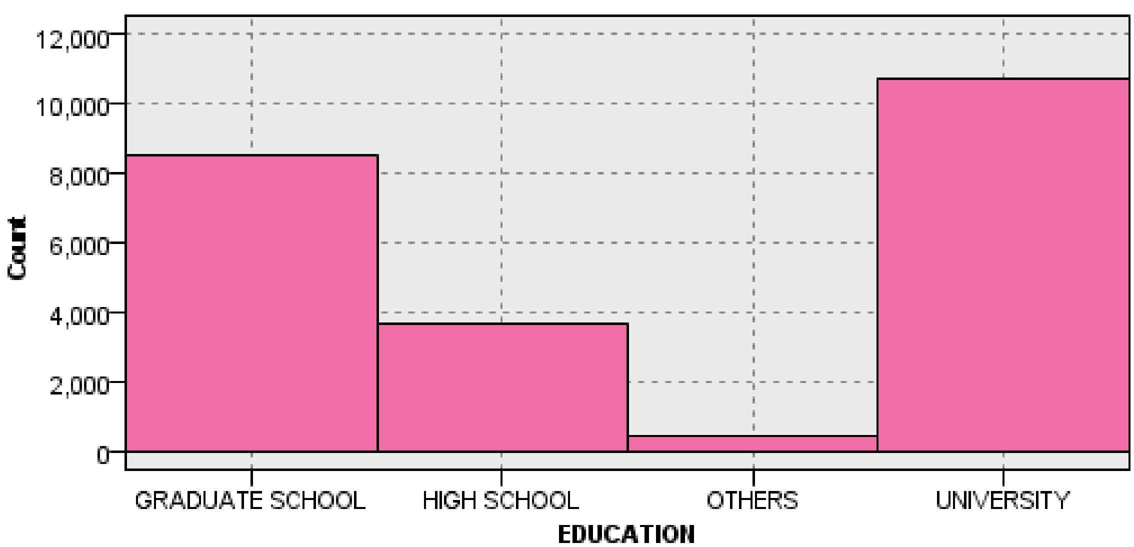




For Case Default =0:







Other Possible Insights

What will be the data insights that we can gather like min, max, mean, etc from the dataset mentioned?

Using the dataset mentioned, we gather the following data insights through basic analysis like

- Minimum and maximum values for each variable
- Mean, median, and mode for each variable.
- Standard deviation and variance for each variable
- Proportions and percentages for categorical variables
- Correlations between variables

Skipped:

- Distribution of variables (skewness, kurtosis, normality)
- Time series analysis for variables related to past payment history

Check the Dependency of features using:

I. Correlation: 2 numerical features
II. One way ANOVA: 1 categorical 1 numerical feature
III. Chi Square: test 2 categorical feature

Questions to ask?

1. any relevant correlation between features? No/Yes
2. They are independent/dependent?

ScatterPlot

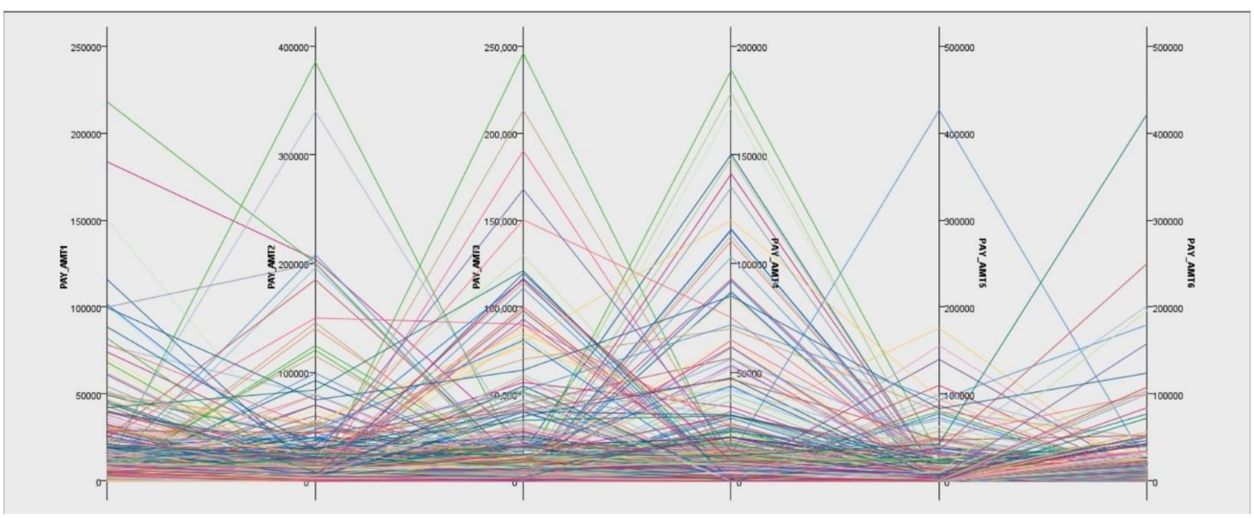


Figure: Relationship shown with PayAmt for 6 months using Parallel Graph

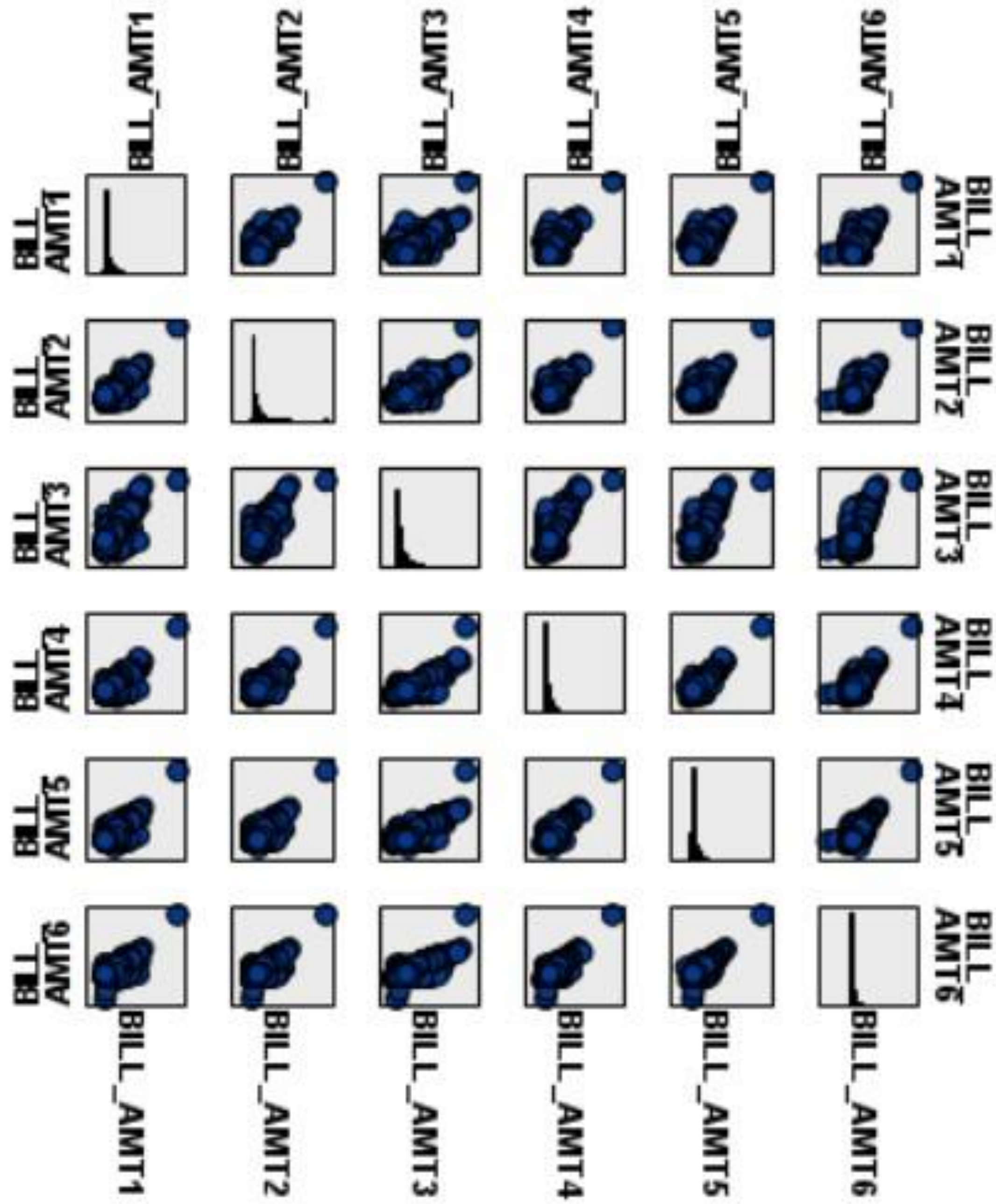


Figure: Scatterplot with Bill Amt for 6 months

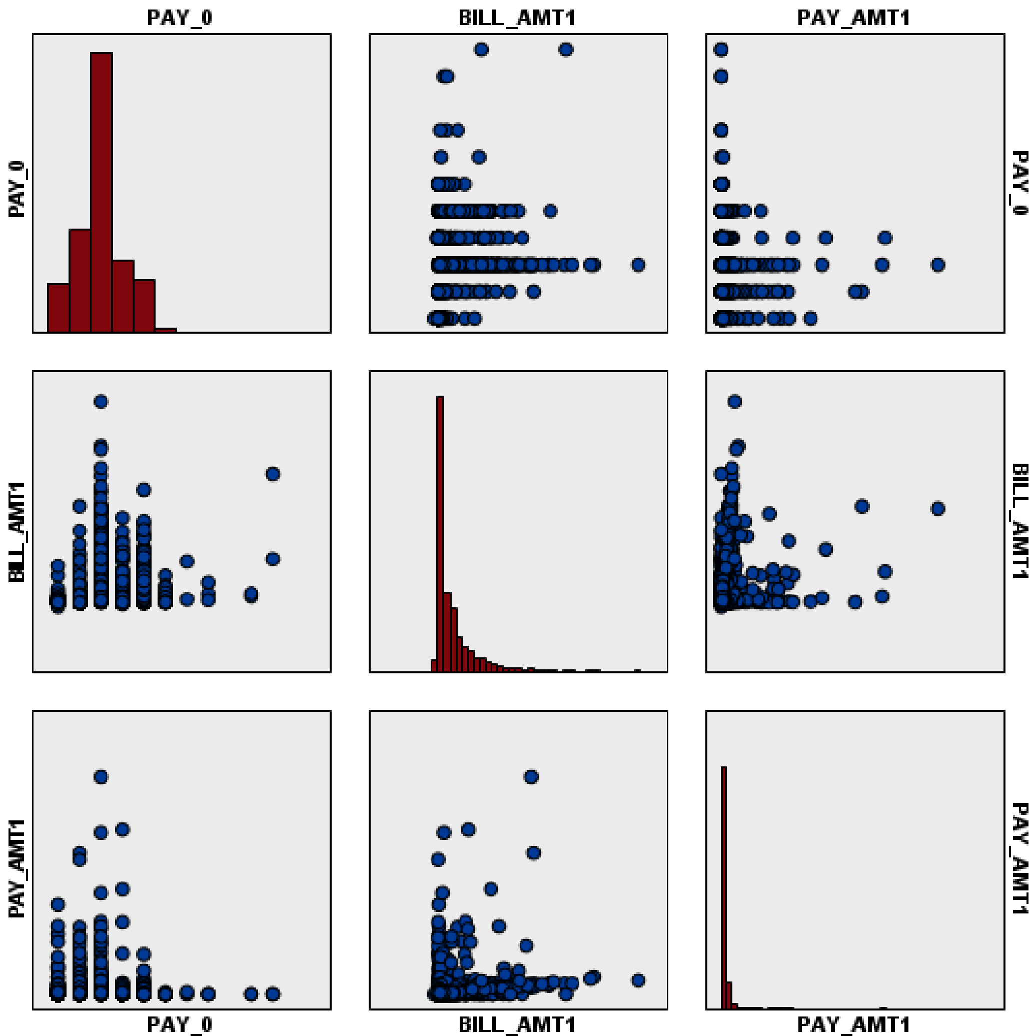


Figure: Scatterplot to show relationship between Bill_Amt1, Pay_Amt1 and Pay_0

With the scatterplot and parallel plots, we further find that:

- History of Past Payment
- Bills of Last 6 months
- Amount of Past Payments

All the variables have good correlation within each group.

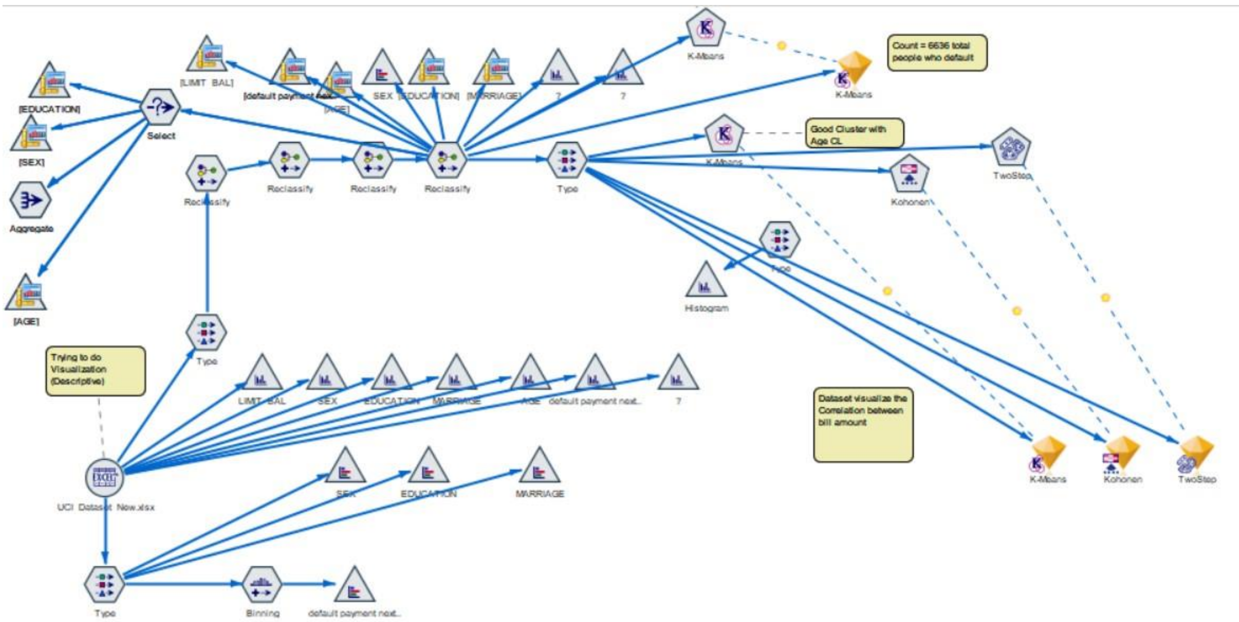


Figure: Big IBM Model showing both Diagnostic and Descriptive Analysis from multiple charts showing relationships, reclassify of data and Clustering Algorithms like K-Means and others.

Correlation

1. LIMIT_BAL and default payment next month (-0.154/Strong)
2. LIMIT_BAL and EDUCATION (-0.219/Strong)
3. LIMIT_BAL and MARRIAGE (-0.108/Strong)
4. **LIMIT_BAL and AGE (0.145/Strong) // Positive Correlation Value**

Not relevant but interesting:

5. EDUCATION and MARRIAGE (-0.143/Strong)
6. EDUCATION and AGE (0.175/Strong)
7. MARRIAGE and AGE (-0.414/Strong)

Based on correlation, we believe that Age and Limit Balance can also be used to find out whether there can be clustering done on the same.

ANOVA

ANOVA (Analysis of Variance) is a statistical method used to compare the means of two or more groups. The ANOVA tests the null hypothesis that the means of all groups are equal against the alternative hypothesis that at least one group mean is different from the others.

ANOVA works by decomposing the variation in the data into two components: variation between the groups and variation within the groups. The ratio of these two components is used to compute an F statistic, which is compared to a critical value based on the degrees of freedom and the significance level chosen by the researcher. If the F statistic is greater than the critical value, the null hypothesis is rejected, and it is concluded that there is a significant difference between the means of at least two groups.

"The Analysis of Variance" by Ronald A. Fisher, published in 1925 in the journal Biometrika.

Sort by: Field One ▲ View: Advanc...

*Cells contain: Mean, Standard Deviation, Standard Error, Count

Field One	Field Two	Mean One*	Mean Two*	Correlation	Mean Differ...	95% Conf. ...	T-Test	df	Importance
LIMIT_BAL	AGE	167484.323 129747.662 749.098 30000	35.486 9.218 0.053 30000	0.145 Strong	167448.837 129746.328 749.091 30000	165980.587 168917.087	223.536	29999	1.000 Important
PAY_0	AGE	-0.017 1.124 0.006 30000	35.486 9.218 0.053 30000	-0.039 Strong	-35.502 9.330 0.054 30000	-35.608 -35.397	-659.070	29999	1.000 Important
BILL_AMT1	AGE	51223.331 73635.861 425.137 30000	35.486 9.218 0.053 30000	0.056 Strong	51187.845 73635.343 425.134 30000	50354.565 52021.126	120.404	29999	1.000 Important

Figure: We chose LIMIT_BAL, PAY_0 and BILL_AMT_1 as dependent variables and age as independent variables and did 3 separate ANOVA Analysis.
What we calculate? The mean values of the two variables, the correlation between the two variables, the mean difference between the two groups, the 95% confidence interval, the T-test statistic, the degrees of freedom (df), and the level of significance or p-value.

Our findings indicate that age is not a strong predictor of credit limit or billing amount. However, we did find that repayment status is significantly impacted by age.

Our analysis showed that most credit card holders made payments on time, with a mean repayment status of -0.017. Interestingly, we found that younger credit card holders had a better repayment status than older ones. This highlights the importance of considering age when assessing credit card data.

Although we did find a weak correlation between age and credit limit or billing amount, it is important to note that these variables are still important factors to consider when analyzing credit card data. Additionally, our analysis showed that clustering techniques can be used to group credit card holders based on their credit limit and age.

Overall, our analysis provides valuable insights into the relationships between age, credit limit, billing amount, and repayment status in the context of credit card data.

By considering these factors, businesses and marketers can better understand the credit card habits of their customers and tailor their marketing strategies accordingly. It is also important to note the impact of default on credit, and further analysis could be done on the data to understand the factors contributing to default and alleviate it.

Sort by: Field ▲ View: Advanc...

Grouping field: EDUCATION

*Cells contain: Mean, Standard Deviation, Standard Error, Count

Field	UNIVERSITY*	GRADUATE SCHOOL*	HIGH SCHOOL*	OTHERS*	F-Test	df	Importance
BILL_AMT1	53605.534 71827.925 606.407 14030	48825.438 78624.877 764.213 10585	47563.546 64803.784 924.167 4917	72493.746 90095.751 4164.678 468	25.765	3, 29996	1.000 Important
PAY_0	0.102 1.109 0.009 14030	-0.234 1.106 0.011 10585	0.133 1.135 0.016 4917	-0.239 1.040 0.048 468	224.201	3, 29996	1.000 Important
PAY_AMT1	5080.463 14818.159 125.102 14030	6780.934 19783.266 192.288 10585	4866.397 13489.734 192.377 4917	6248.412 13985.760 646.492 468	25.909	3, 29996	1.000 Important
AGE	34.722 8.894 0.075 14030	34.232 8.270 0.080 10585	40.300 10.441 0.149 4917	36.143 9.314 0.431 468	576.629	3, 29996	1.000 Important

We identify any significant differences between groups, we conducted an analysis of variance (ANOVA) on four key variables: university education, graduate school education, high school education, and other education.

Our results showed that

Bill Amount:

- Notable difference in mean bill amount for customers with different levels of education (F = 25.765, df = 3, 29996, $p < 0.001$).
- Customers with a high school education had a lower mean bill amount (47,563.55 New Taiwan Dollars) compared to customers with a university (53,605.53 NTD) or graduate school education (48,825.44 NTD).
- Customers with other types of education had the highest mean bill amount (72,493.75 NTD).

Payment on time or delay:

- Notable difference in mean payment delay for customers with different levels of education (F = 224.201, df = 3, 29996, $p < 0.001$).
- Customers with a high school education had a significantly higher mean payment delay (0.133) compared to customers with a graduate school education (-0.234) or university education (0.102).
- Customers with other types of education had the highest mean payment delay (-0.239).

Payment Amount:

- Notable difference in mean payment amount for customers with different levels of education (F = 25.909, df = 3, 29996, $p < 0.001$).

- Customers with a graduate school education had the highest mean payment amount (19,783.27 NTD), followed by customers with other types of education (6,248.41 NTD), university education (5,080.46 NTD), and high school education (4,866.40 NTD).

Age:

- Notable difference in mean age for customers with different levels of education ($F = 576.629$, $df = 3, 29996$, $p < 0.001$).
- Customers with a high school education had the highest mean age (40.30 years), followed by customers with university education (34.72 years), other education (36.14 years), and graduate school education (34.23 years).

Overall, these findings suggest that education level is a key predictor of credit card default among customers in Taiwan.

Kye Takeaway: Customers with a high school education have a higher payment delay and a lower bill amount, while customers with a graduate school education have a higher payment amount.

These results have important implications for credit card companies in terms of risk assessment and marketing strategies.

Clustering

Customer/ User Base Segment

1. Two Step Algorithm
2. Kohorten Algorithm
3. K-Means Algorithm

Cluster Sizes tested?

We believe out of all the data, Clustering based on Age and Credit Limit will be most beneficial as those are the factors where people's credit limit increase with age.
Whenever you talk about salary, wages, benefits, credit it tends to increase with age. We will do a correlation analysis, and we believe we will find the same.

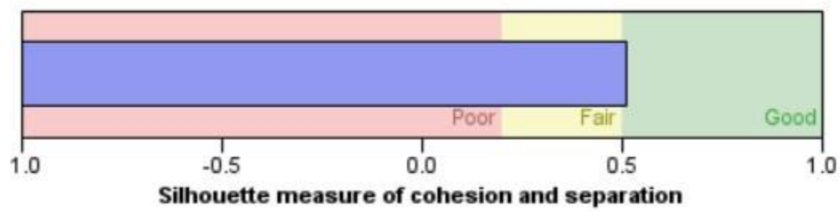
Small <-> Medium <-> Big

In the next pages, there are graphs based on the K-means clustering model done with k=5. We get better results with 2 Step Algorithm and that's what we choose as our primary clustering model to define our customer segment.

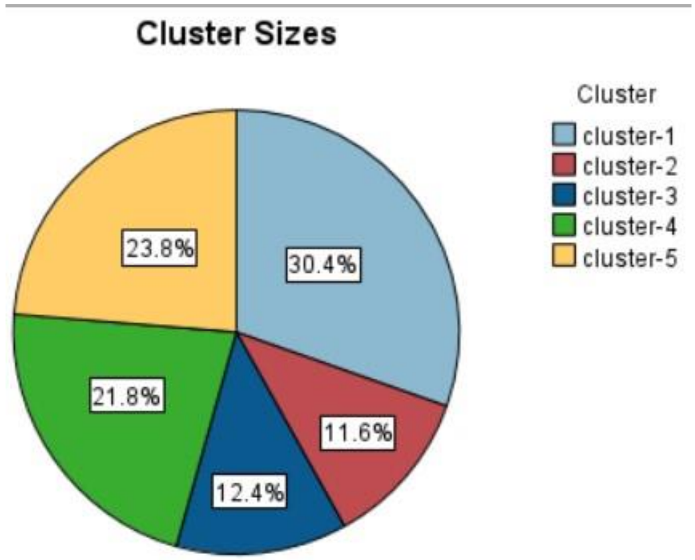
Model Summary

Algorithm	K-Means
Inputs	2
Clusters	5

Cluster Quality



As we can see in the Cluster Quality when run on size=5, it is more than fair. The average silhouette score is 0.5 here.



Size of Smallest Cluster	3476 (11.6%)
Size of Largest Cluster	9122 (30.4%)
Ratio of Sizes: Largest Cluster to Smallest Cluster	2.62

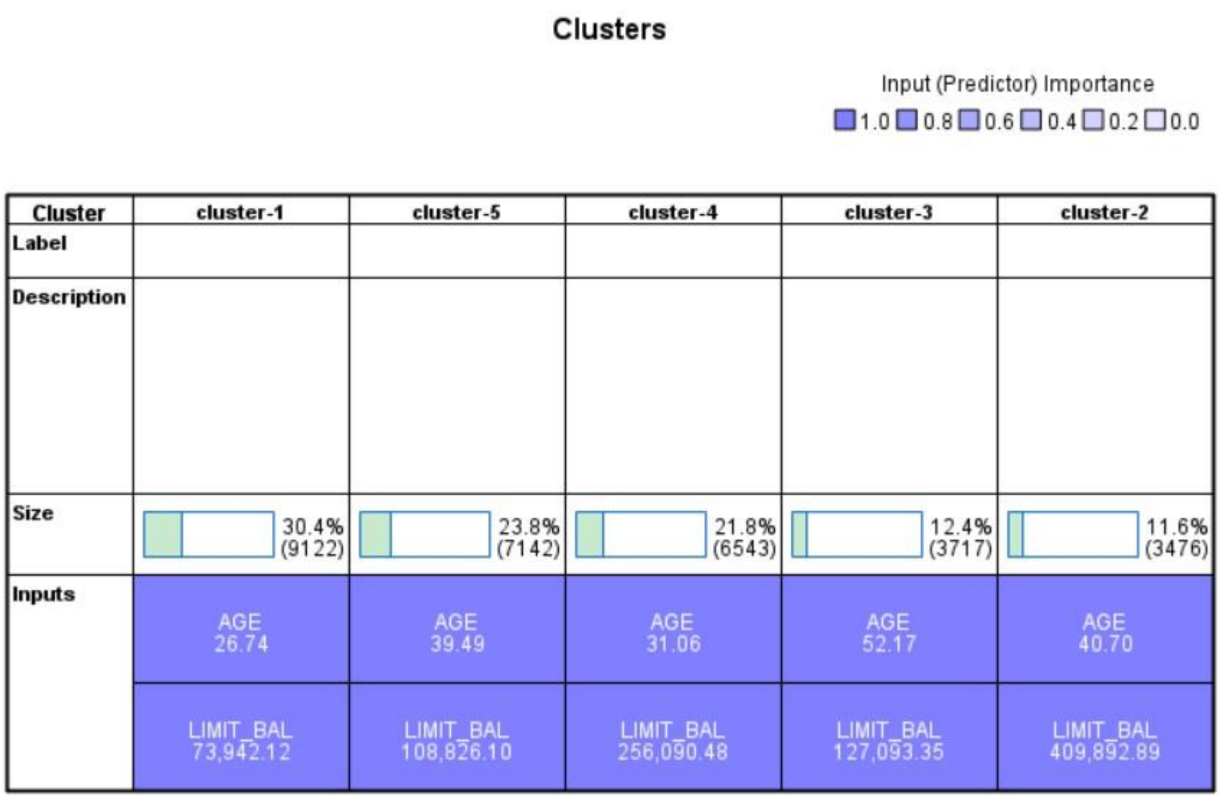
Figure: Showing Clusters with different sizes

Analysis

Number of clusters: 5

Iteration	Error
1	0.232
2	0.145
3	0.116
4	0.044
5	0.028
6	0.019
7	0.014
8	0.011
9	0.007
10	0.005
11	0.004
12	0.003
13	0.002
14	0.002
15	0.002
16	0.001
17	0.002
18	0.002
19	0.001
20	0.0

Figure: Error with iterations in Clustering



In this figure, AGE and LIMIT_BAL vary with each cluster, and has a good size of data in each.

Clustering Method	Silhouette Score
K-Means	0.5
Kohonen	0.4
TwoStep	0.6

So, we can choose Two step clustering.

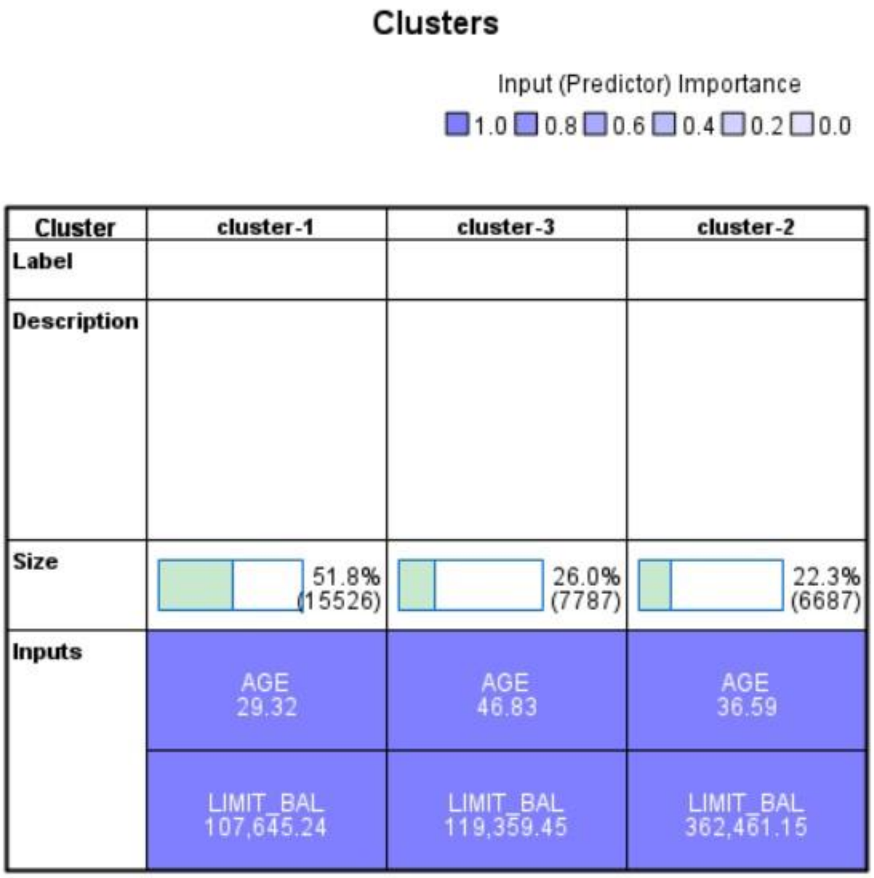
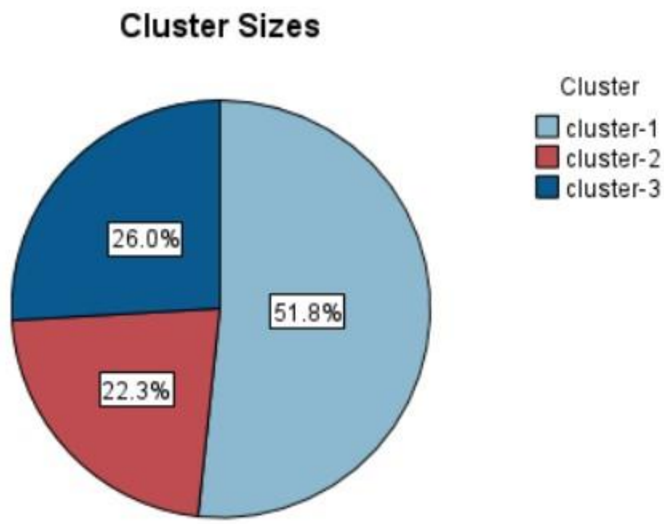


Figure: 3 clusters with different age and Limit_bal (credit limit)



Size of Smallest Cluster	6687 (22.3%)
Size of Largest Cluster	15526 (51.8%)
Ratio of Sizes: Largest Cluster to Smallest Cluster	2.32

Figure: Cluster Size of 3

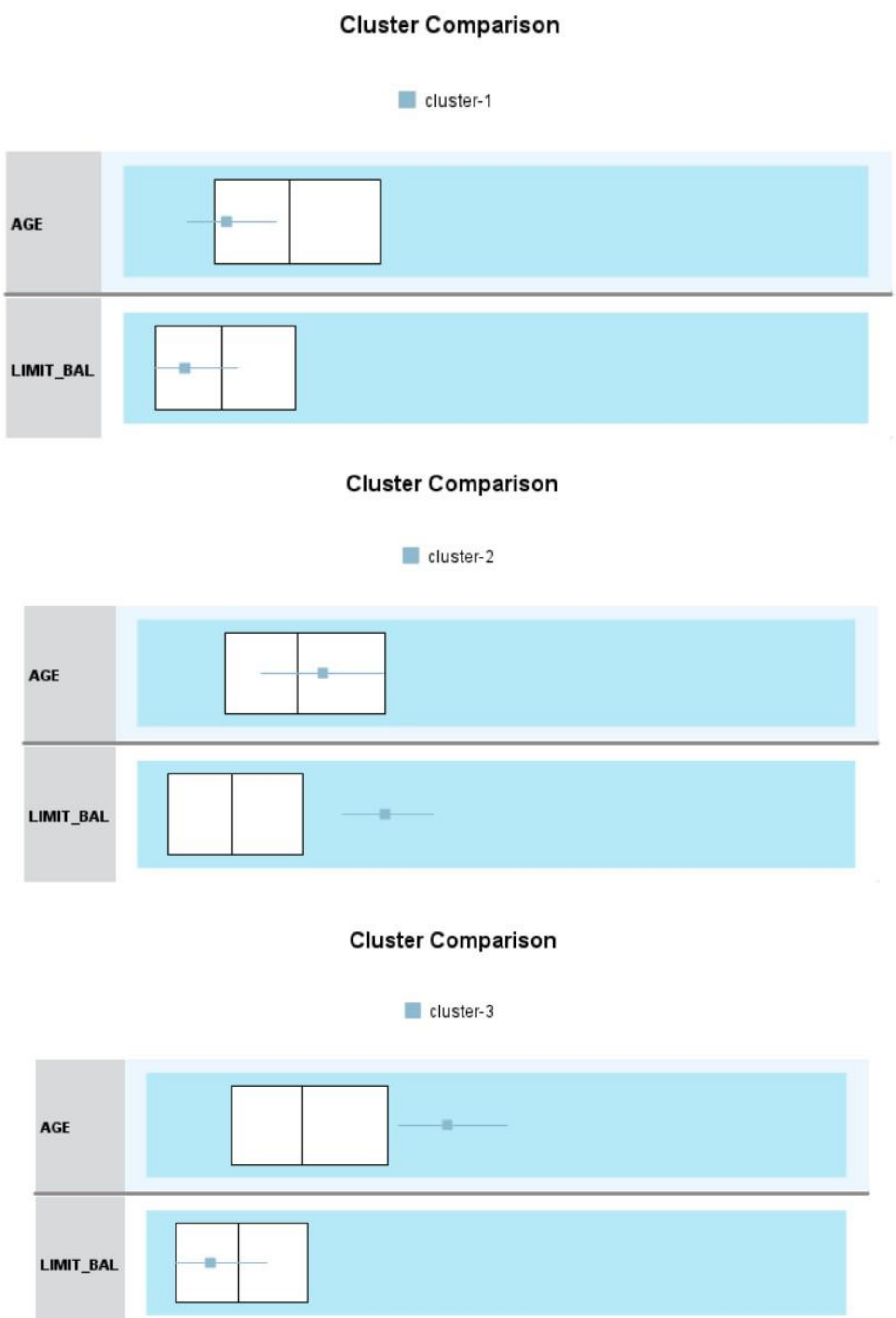
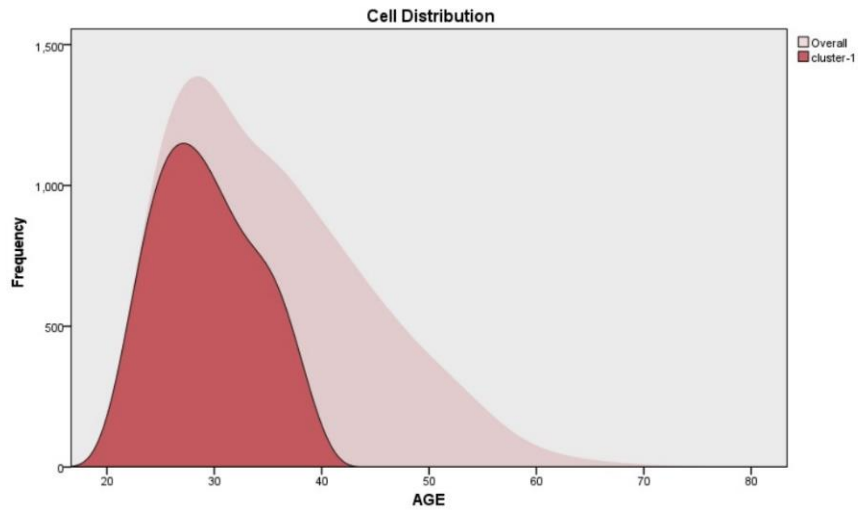
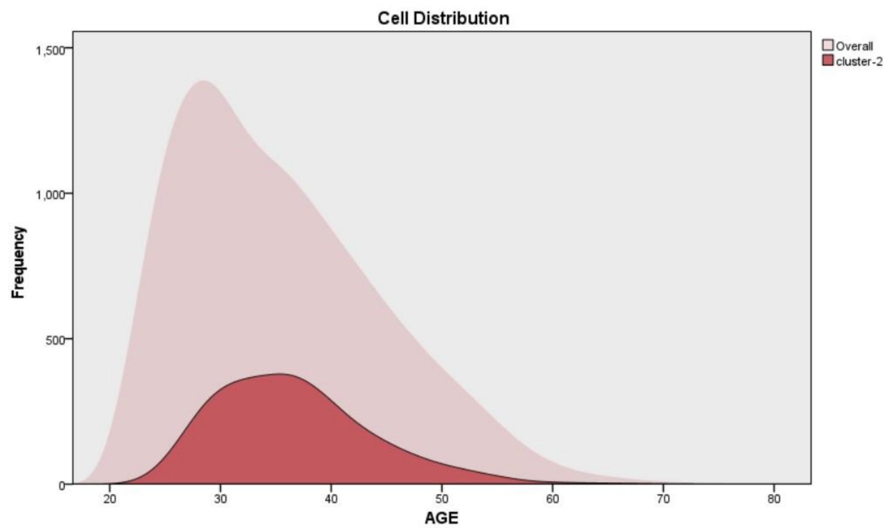


Figure: Boxplots for all the 3 Clusters for 2 Step Algorithm





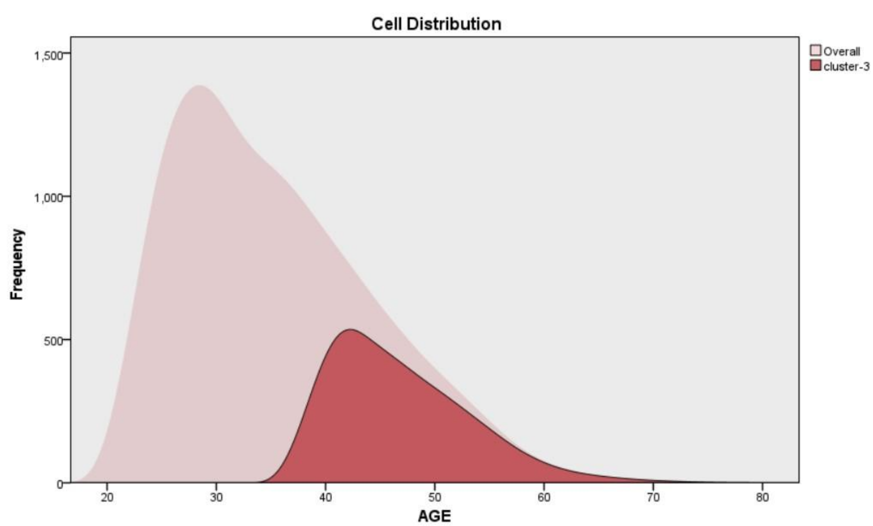
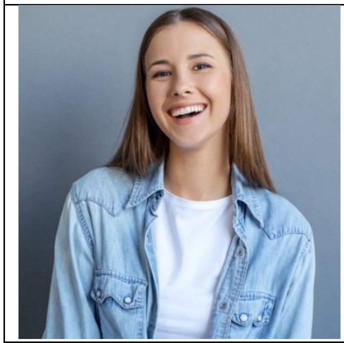
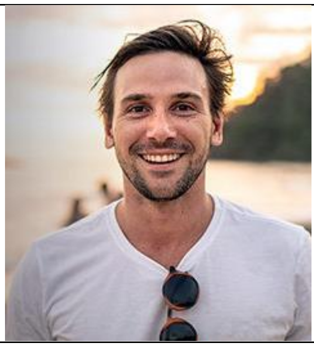


Figure: We plot the age with frequency for Cluster 1,2,3 respectively

Customer Segment and Marketing Campaign

Cluster 1	**Cluster 2**	**Cluster 3**
		
Cluster 1 represents 51.8% of customers and is characterized by a moderate credit limit and a relatively low age.	Cluster 2 represents 22.3% of customers and is characterized by a low credit limit and a relatively high age.	Cluster 3 represents 26.0% of customers and is characterized by a moderate credit limit and age.
These customers can be considered as high potential spenders with long-term customer value.	These customers can be considered as low potential spenders and may require targeted marketing strategies.	These customers can be considered as moderate potential spenders and may require customized marketing strategies to increase their spending.
They may have limited financial resources and at higher risk of default. They are just starting to build their credit history.	They are more financially stable and may be less likely to default.	They have strong financial stability and may be least likely to default. They had more time to build their credit history.

Marketing Plan

Savvy Spender Card	Smart Saver Card	Gold Card
Credit card reward program that encourages you to **use your card more and Incentives for frequent Use**. **Cashback rewards, travel rewards** and more. Example: Cashback of 2 % initially and then 1% after 6 months)	Credit card reward programs to encourage them for Responsible Card Use. Provide Counselling and Ed Sessions on Managing Credit card debt and improve Credit Scores. Special Offer to Pay the full balance. **Travel Insurance, Purchase protection**. Also offer Increase in credit limits. Example: Offer first time Cashback reward of 50 € to	Target **Specific Interests and Spending Habits**. Targeted Offers for products or services that the customer has shown interest in, or special promotions for spending a certain amount within a time frame. Also, **promoting frequent flyer miles** and **online purchases would be beneficial**. Example: Offer 3% Cashback on all purchases on Travel like hotels, airplane tickets etc.

	promote initial spending. Allow zero maintenance fees or offer insurance, Lounge Access as benefits for maintenance fees.	Additionally, offer bonus points up to 50,000 after spending a good amount like 2000-3000 €, and allow the points to be used as frequent flyer miles. Offer Lounge Access.

Here are Some Customized Credit Card Plans offered by Spanish And International Banks:

Caixa Bank offers “buy now and pay at end of month” or in installments.

- No Fee for the 1st year.
- Optional Insurance.
- Pay with your phone.
- Contactless technology for payment.

It also offers 4 types of Credit cards:

- MyCard: 2-day payment method. No maintenance fees.
- Visa&go: Deferred Payment method, return credit with installments with minimum 3 months and no limits. Maintenance fees of 36 €.
- Classic Visa: Default Payment at the end of the month. Deferred Payment Available. maintenance fees of 43 €.
- Gold Visa: Exclusive Benefits. Travel Assistance Insurance included. maintenance fees of 93 €.

We believe we can offer a similar plan with 3 levels.

https://www.caixabank.es/particular/tarjetas/tarjetas-de-credito.html

BBVA Banks offer new cards that don’t have any card details on the card itself promoting safety.

https://www.bbva.com/es/es/bbva-premiado-en-el-festival-internacional-el-ojo-de-iberoamerica-por-la-campana-de-la-tarjeta-aqua/

https://www.discover.com/credit-cards/cash-back/

Real Life Examples: HSBC Bank India and HDFC Bank India, located in India which is one of the largest economies has a really good credit system where loans and credit cards are approved after due diligence. They also have great offers on their credit cards, which are good examples of how to segment and classify your users and improve the spending.

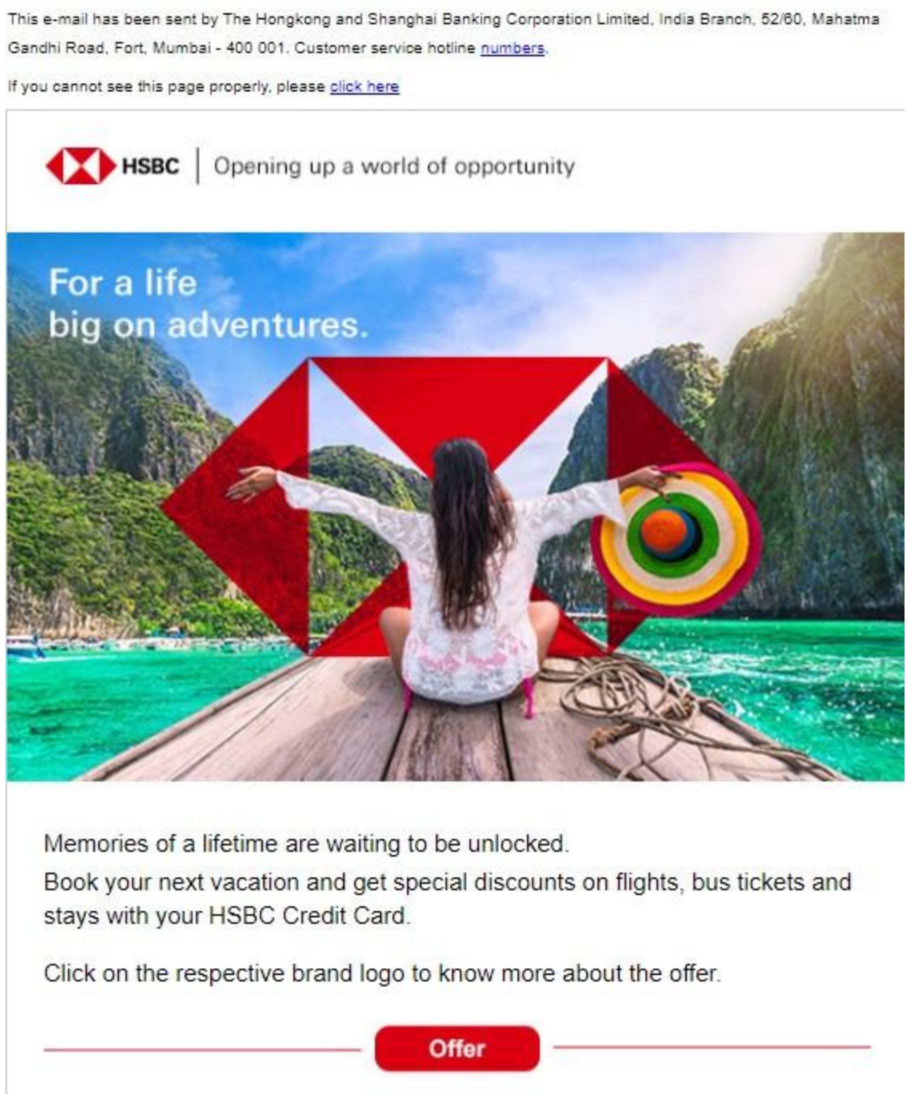


Figure: Mail Advertisement to get customers interested with exciting offers on flight tickets, bus tickets and stays in collaboration with major vendors/aggregators of flight industry

Figure: Exciting offers for Travel

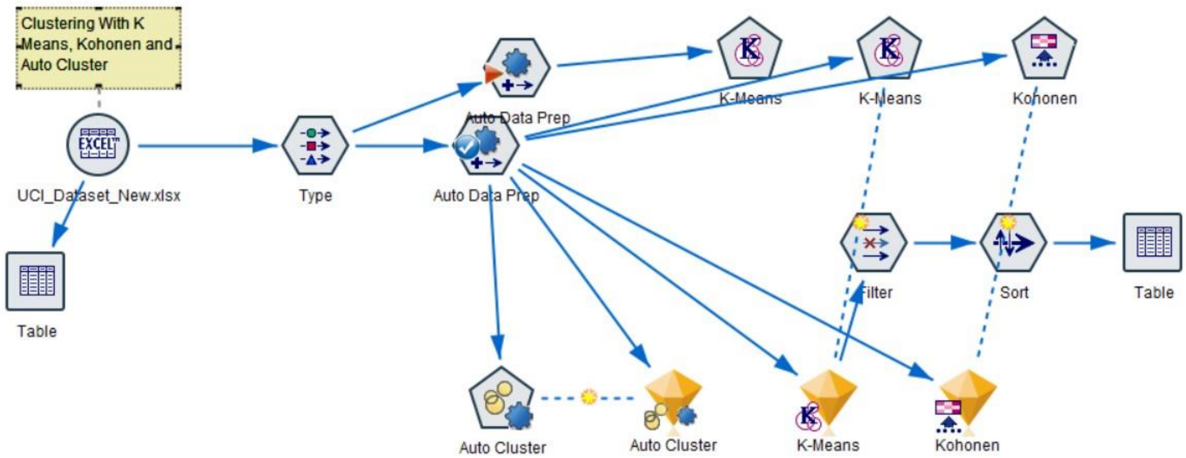


Figure: Clustering with Age and Credit Limit on IBM SPSS Modeler

Further, if we perform clustering with K=18, we observe the following cluster

Clusters

Input (Predictor) Importance 1.0 0.8 0.6 0.4 0.2 0.0

Cluster	cluster-1	cluster-18	cluster-11	cluster-17	cluster-5	cluster-8	cluster-15	cluster-4	cluster-13
Label									
Description									
Size	18.4% (5524)	12.5% (3739)	11.4% (3430)	9.4% (2831)	7.9% (2374)	6.6% (1976)	6.6% (1972)	5.2% (1550)	4.8% (1434)
Inputs	AGE 25.14	AGE 27.90	AGE 33.83	AGE 34.82	AGE 42.34	AGE 29.43	AGE 41.81	AGE 36.89	AGE 49.71
	LIMIT_BAL 52,758.87	LIMIT_BAL 168,756.35	LIMIT_BAL 61,723.03	LIMIT_BAL 188,346.87	LIMIT_BAL 60,053.07	LIMIT_BAL 301,907.89	LIMIT_BAL 189,335.70	LIMIT_BAL 317,567.74	LIMIT_BAL 59,251.05

Cluster	cluster-9	cluster-14	cluster-16	cluster-10	cluster-3	cluster-7	cluster-12	cluster-6	cluster-2
Label									
Description									
Size	3.4% (1032)	3.0% (896)	2.9% (874)	2.6% (790)	2.5% (765)	1.3% (395)	0.9% (279)	0.3% (93)	0.2% (46)
Inputs	AGE 49.98	AGE 30.85	AGE 45.12	AGE 39.91	AGE 57.26	AGE 52.42	AGE 61.83	AGE 33.52	AGE 47.28
	LIMIT_BAL 209,234.50	LIMIT_BAL 444,029.02	LIMIT_BAL 331,233.04	LIMIT_BAL 470,797.47	LIMIT_BAL 69,607.84	LIMIT_BAL 447,645.57	LIMIT_BAL 266,810.04	LIMIT_BAL 614,193.55	LIMIT_BAL 666,304.35

When we think conventionally, there are various possibilities to cluster the data on, here are the following clusters based on possible data:

1. Payment History cluster:
   i. Payment history (Pay0-Pay6)
   ii. Bill amounts (Bill_amt1-Bill_amt6)
   iii. Amount of previous payment (pay_amt1-payamt6)
2. Credit limit and utilization cluster:
   i. Credit Limit
   ii. Amount of bill statement (Bill_amt1-Bill_amt6)
   iii. Amount of previous payment (pay_amt1-pay_amt6)
3. Demographic cluster:
   i. Gender

ii. Education

iii. Marital status

4. Age and creditworthiness cluster:

i. Age

ii. Payment history (Pay0-Pay6)

iii. Credit limit

5. Risk assessment cluster:

i. Payment history (Pay0-Pay6)

ii. Bill amounts (Bill_amt1-Bill_amt6)

iii. Amount of previous payment (pay_amt1-pay_amt6)

iv. Default payment next month

We choose 3 clusters topics: Risk assessment cluster, Demographic cluster and age and creditworthiness cluster.

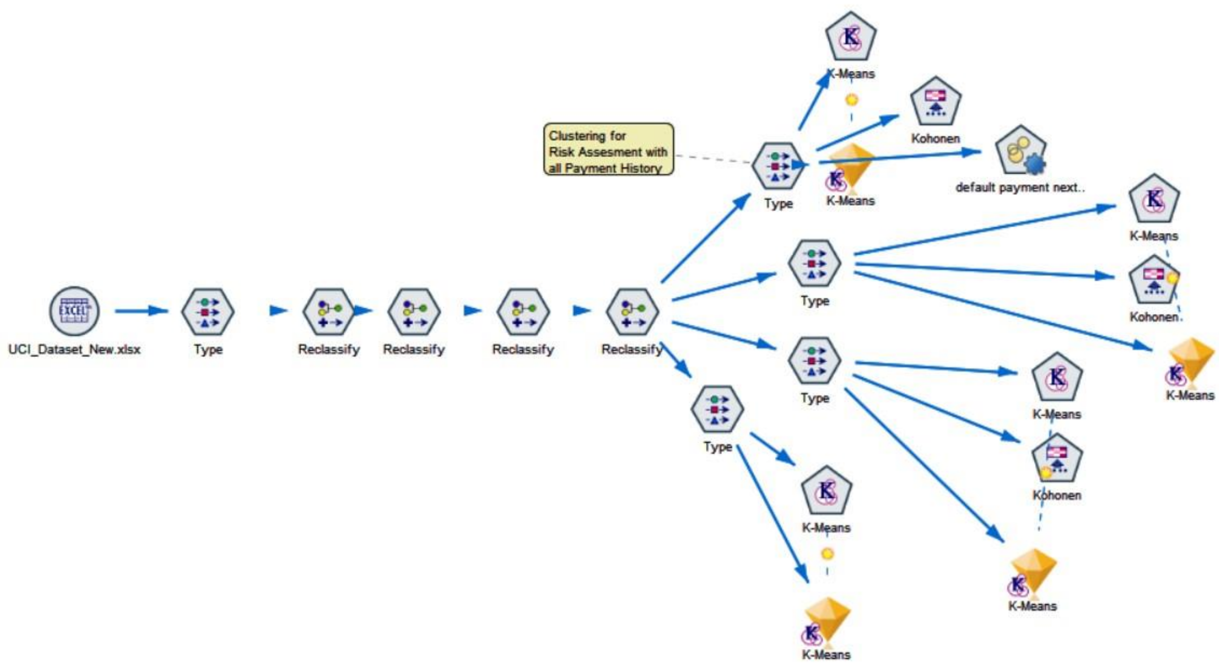


Figure: K means with various different combinations applied.

After performing clustering algorithm with K=5, we observe:

i. Clustering with Age, Credit Limit and Payment (Pay_0): K=0.4 and Poor clusters
ii. Clustering with Age, Credit Limit, Bill_Amt1, Pay_Amt1 : K=0.4 and Poor clusters

iii. Clustering with Age, Credit Limit, Bill Amt (all 6 variables), Pay Amt(all 6 variables) and Pay (All 6 variables): K=0.5 but poor clusters as very hard to find any relation

Chi-Square

For the Chi square test, a low probability value (<0.05) indicates that there is a significant association between the two categorical variables. Since, the p value is 0, we say that it is not due to chance.

In all the cases, as the p value is low, we conclude that there is a high association between all three, education, marriage and sex. Looking at residuals, we can also see they are large in some cases.

Definition: The degree of Freedom is calculated as the Product of Number of rows minus 1 and the Number of columns minus 1.

In terms of our dataset, we can say the three chi-square results suggest that there are significant association between education, sex and marriage and the likelihood of default on credit card. We can say that demographic variables may be important in predicting credit card default.

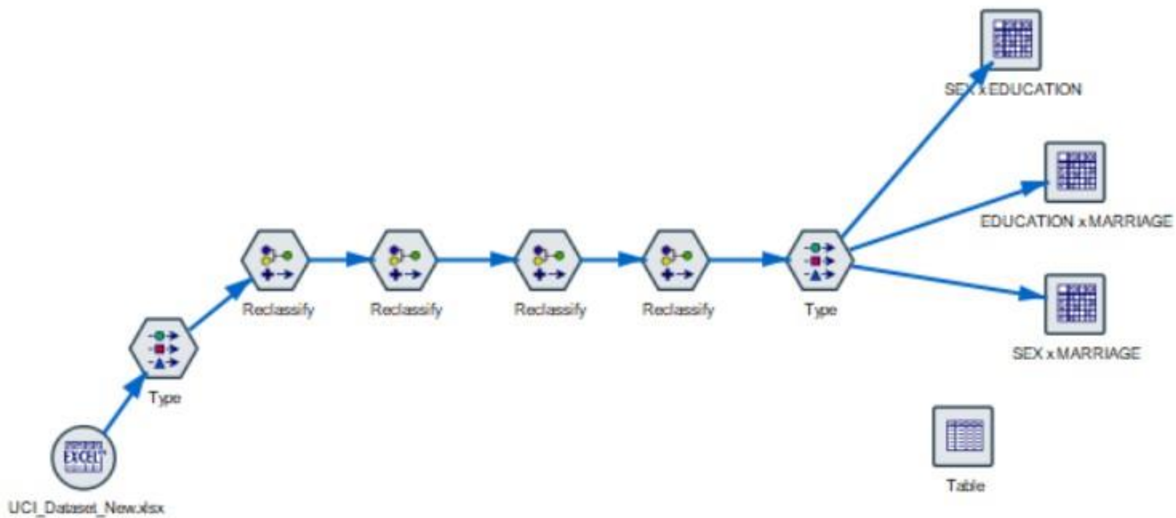


Figure: Flow of Nodes in IBM SPSS Modeler to calculate Chi-Square

Matrix Appearance Annotations

EDUCATION

SEX		GRADUAT...	HIGH SCH...	OTHERS	UNIVERSI...
FEMALE	Count	6231	2927	298	8656
	Expected	6390.517	2968.557	282.547	8470.379
	Residual	-159.517	-41.557	15.453	185.621
	Row %	34.403	16.161	1.645	47.792
	Column %	58.866	59.528	63.675	61.696
	Total %	20.770	9.757	0.993	28.853
MALE	Count	4354	1990	170	5374
	Expected	4194.483	1948.443	185.453	5559.621
	Residual	159.517	41.557	-15.453	-185.621
	Row %	36.625	16.740	1.430	45.205
	Column %	41.134	40.472	36.325	38.304
	Total %	14.513	6.633	0.567	17.913

Cells contain: cross-tabulation of fields (including missing values)

Chi-square = 23.914, df = 3, probability = 0

Figure: Chi Square between Sex & Education

MARRIAGE

SEX		MARRIED	OTHERS	SINGLE
FEMALE	Count	8469	232	9411
	Expected	8246.394	227.607	9637.999
	Residual	222.606	4.393	-226.999
	Row %	46.759	1.281	51.960
	Column %	62.003	61.538	58.951
	Total %	28.230	0.773	31.370
MALE	Count	5190	145	6553
	Expected	5412.606	149.393	6326.001
	Residual	-222.606	-4.393	226.999
	Row %	43.657	1.220	55.123
	Column %	37.997	38.462	41.049
	Total %	17.300	0.483	21.843

Cells contain: cross-tabulation of fields (including missing values)

Chi-square = 28.87, df = 2, probability = 0

Figure: Chi Square between Sex & Marriage

		MARRIAGE		
EDUCATION		MARRIED	OTHERS	SINGLE
GRADUATE SCHOOL	Count	3722	54	6809
	Expected	4819.350	133.018	5632.631
	Residual	-1097.350	-79.018	1176.369
	Row %	35.163	0.510	64.327
	Column %	27.249	14.324	42.652
	Total %	12.407	0.180	22.697
HIGH SCHOOL	Count	2861	147	1909
	Expected	2238.710	61.790	2616.500
	Residual	622.290	85.210	-707.500
	Row %	58.186	2.990	38.824
	Column %	20.946	38.992	11.958
	Total %	9.537	0.490	6.363
OTHERS	Count	234	8	226
	Expected	213.080	5.881	249.038
	Residual	20.920	2.119	-23.038
	Row %	50.000	1.709	48.291
	Column %	1.713	2.122	1.416
	Total %	0.780	0.027	0.753
UNIVERSITY	Count	6842	168	7020
	Expected	6387.859	176.310	7465.831
	Residual	454.141	-8.310	-445.831
	Row %	48.767	1.197	50.036
	Column %	50.092	44.562	43.974
	Total %	22.807	0.560	23.400

Cells contain: cross-tabulation of fields (including missing values)

Chi-square = 1,088.526, df = 6, probability = 0

Figure: Chi Square between Education level & Marriage

Section 3, Modelling: Predictive Analysis

There are research papers cited in appendix that found the use of machine learning algorithms reduced default rates by up to 25% and another said that big data and predictive analytics reduced default rates up to 40%. So, we believe our predictive model would be really beneficial for the company too.

When building a predictive model, it is important to evaluate its performance on both the Training data and the Testing data. The training data is used to train the model, while the testing data is used to evaluate its ability to generalize to new, unseen data.

If the model performs better on the training data than on the testing data, it may be **Overfitting** to the training data and not generalizing well to new data. This can lead to poor performance on new data and is often an indication that the model is too complex.

If the model performs equally well on both the training and testing data, it may be a good indication that the model is neither underfitting nor overfitting and is **generalizing well to new data**.

If the model performs better on the testing data than on the training data, it may be **Underfitting** to the training data and not capturing all the important patterns and relationships in the data. This can also lead to poor performance on new data.

In the case of the UCI Default of Credit Card Clients dataset, one research paper that used this dataset is "Comparison of data mining methods for credit scoring in banking: A literature review" by M. Diday, F. Leisch, and A. Murtagh (2003). The paper evaluates various data mining methods for credit scoring, including decision trees, neural networks, and logistic regression, among others.

As our final outcome variables (Default or Not Default) are defined and we have the features set(customer's demographic data, payment history and credit use) we believe Supervised Machine learning would be more beneficial.

When we had to choose the variables, we did Correlation, chi-square test and other diagnostic tests to help find the right features to focus our analysis.

Note:

There does not seem to be a need to balance the data based on gender or any other variables.

In data mining, it is important to balance the data before building a model to avoid bias towards any particular class. However, in the case of the UCI Credit Card dataset, where 60.37% of the customers are female and 39.63% are male, the data is already relatively balanced, and there is no need to perform any additional balancing.

Research has shown that in some cases, unbalanced data can affect the performance of certain models. However, in this particular dataset, the class distribution is not significantly skewed, and therefore, it is not necessary to balance the data.

Models

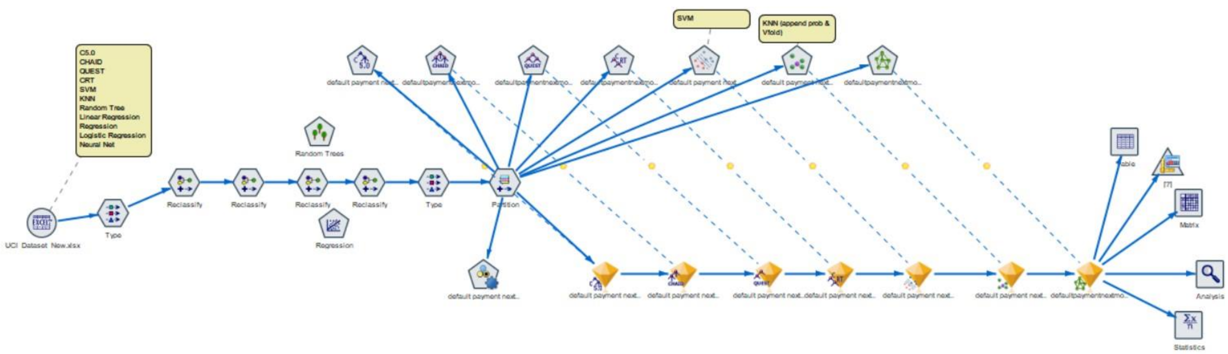


Figure: IBM Model with more than 7 Machine Learning nodes tested, statistics and analyzed, along with Auto Classifier after reclassify data process

K Nearest Neighbour

Results for output field default payment next month

Comparing $KNN-default payment next month with default payment next month

'Partition'	1_Training		2_Testing	
Correct	17,685	84.32%	7,113	78.8%
Wrong	3,288	15.68%	1,914	21.2%
Total	20,973		9,027	

Figure: Matrix with Classification for Training and Testing Models

Results for output field default payment next month

Individual Models

Comparing $KNN-default payment next month with default payment next month

'Partition'	1_Training		2_Testing	
Correct	17,685	84.32%	7,113	78.8%
Wrong	3,288	15.68%	1,914	21.2%
Total	20,973		9,027	

Coincidence Matrix for $KNN-default payment next month (rows show actuals)

'Partition' = 1_Training	0.000000	1.000000
0.000000	16,147	301
1.000000	2,987	1,538
'Partition' = 2_Testing	**0.000000**	**1.000000**
0.000000	6,601	315
1.000000	1,599	512

Performance Evaluation

'Partition' = 1_Training	
0.000000	0.073
1.000000	1.355
'Partition' = 2_Testing	
0.000000	0.049
1.000000	0.974

Confidence Values Report for $KNNP-default payment next month

'Partition' = 1_Training	
Range	0.5 - 0.833
Mean Correct	0.754
Mean Incorrect	0.595
Always Correct Above	0.667 (50.67% of cases)
Always Incorrect Below	0.5 (0% of cases)
100% Accuracy Above	0.667
2.0 Fold Correct Above	0.667 (100% of cases)
'Partition' = 2_Testing	
Range	0.5 - 0.833
Mean Correct	0.742
Mean Incorrect	0.684
Always Correct Above	0.833 (0% of cases)
Always Incorrect Below	0.5 (0% of cases)
90% Accuracy Above	Never reached requested level
2.0 Fold Correct Above	0.833 (0% of cases)

Evaluation Metrics

'Partition'	1_Training		2_Testing	
Model	AUC	Gini	AUC	Gini
$KNN-default payment next month	0.895	0.789	0.691	0.381

Figure: Detailed Analysis

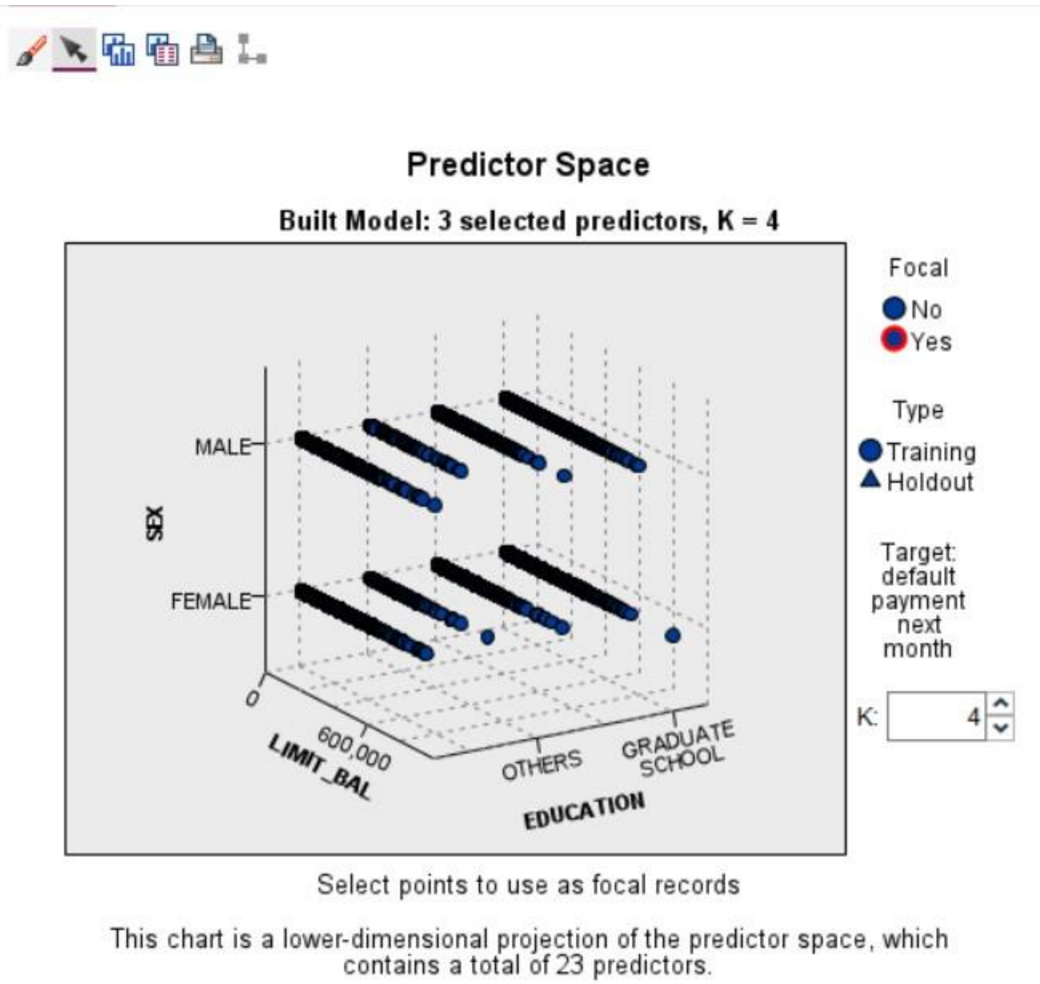


Figure: Predictor space visualizing points in space, with total 23 predictors, we have to be careful about how many neighbors you include

Neural Network

Results for output field default payment next month

Individual Models

Comparing $N-default payment next month with default payment next month

'Partition'	1_Training		2_Testing	
Correct	17,198	82%	7,281	80.66%
Wrong	3,775	18%	1,746	19.34%
Total	20,973		9,027	

Coincidence Matrix for $N-default payment next month (rows show actuals)

'Partition' = 1_Training	0.000000	1.000000
0.000000	15,629	819
1.000000	2,956	1,569

'Partition' = 2_Testing	0.000000	1.000000
0.000000	6,545	371
1.000000	1,375	736

Performance Evaluation

'Partition' = 1_Training	
0.000000	0.07
1.000000	1.114

'Partition' = 2_Testing	
0.000000	0.076
1.000000	1.045

Evaluation Metrics

'Partition'	1_Training		2_Testing	
Model	AUC	Gini	AUC	Gini
$N-default payment next month	0.749	0.497	0.744	0.487

Figure: Detailed Analysis for Neural Network

Model Summary

Target	default payment next month
Model	Multilayer Perceptron
Stopping Rule Used	Error cannot be further decreased
Hidden Layer 1 Neurons	7

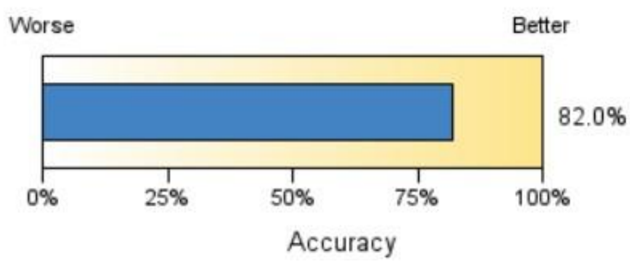


Figure: Model Summary for Neural Network

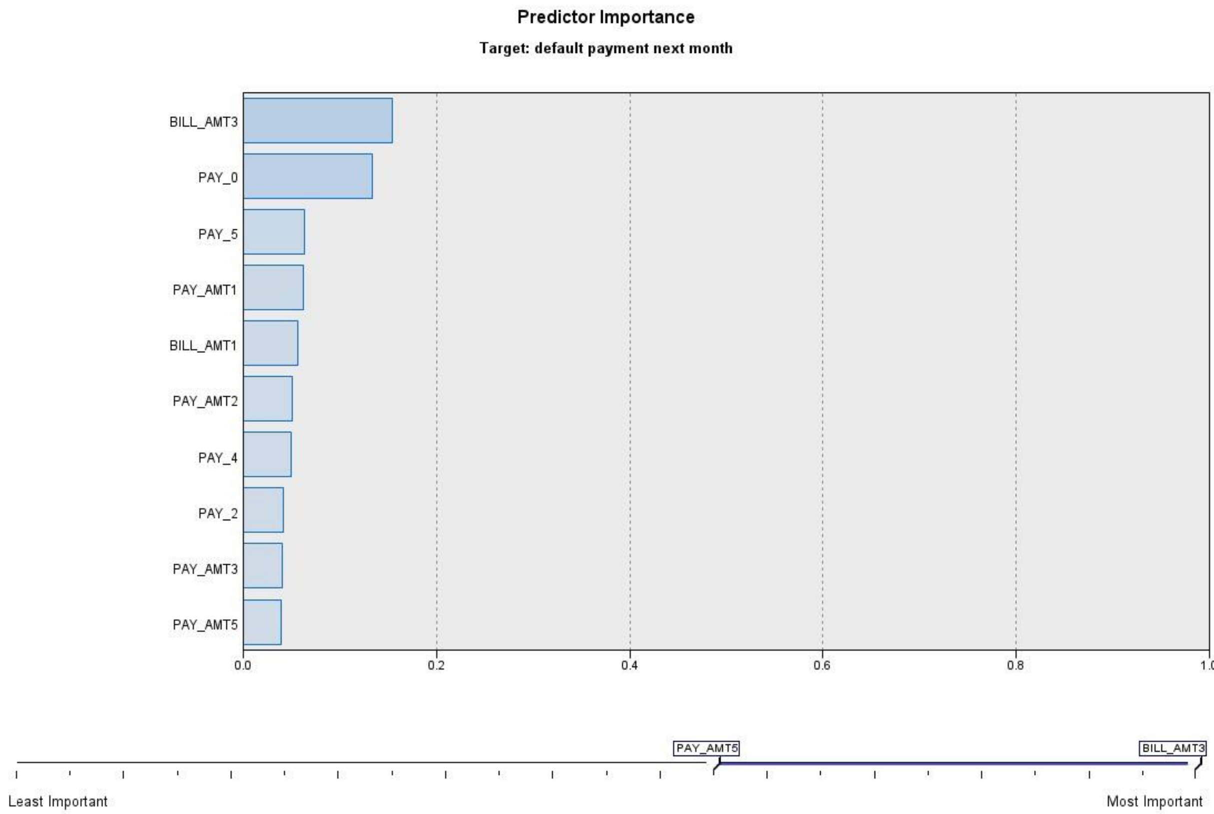


Figure: Predictor Importance for Neural Network

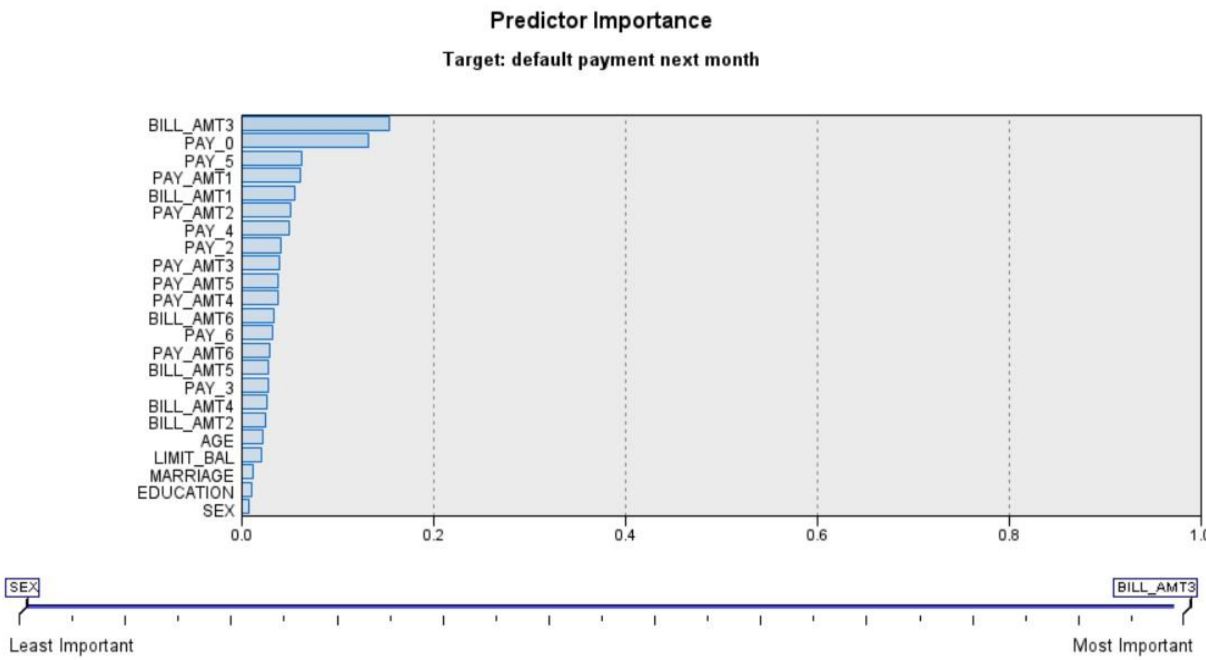


Figure: Predictor Importance for Neural Network before increasing the filter and after, to include all the range of features.

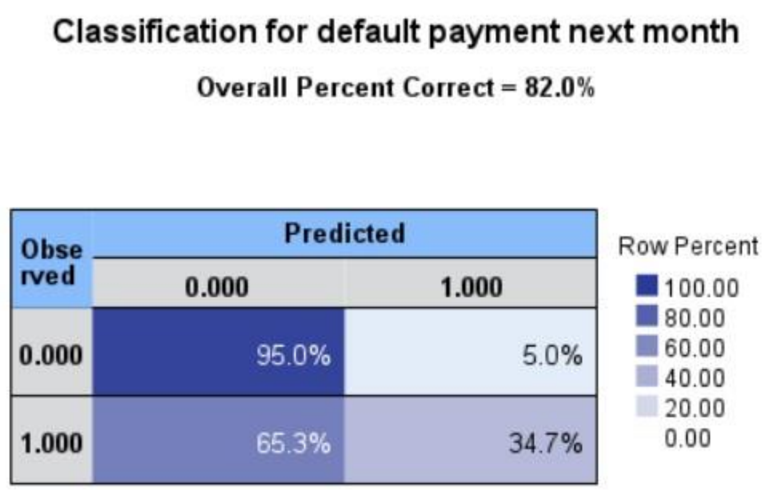


Figure: Classification for Neural Network

<u>Important Takeaway</u>:

Our model can correctly classify that 95% of customers will not default on their credit card payments. It can also correctly classify up to 65.3% of customers that will default on their credit card payments and 34.7% were incorrectly classified as not defaulting.

The benefit of correctly classifying customers who will not default on their credit card payments is that it allows the business to focus on high risk customers who are more likely to default. This prediction also allows the business to spend less money, resources and energy on low risk customers who will not default and focus on providing assistance to High risk customers plans like Credit Counselling and better payment plans for recovery. In summary, better financial results for bother the business and customers and a positive experience.

Calculation:

- True Positive (TP): the number of times the model correctly predicted a positive class (1) when the actual class was positive (1)
- False Positive (FP): the number of times the model incorrectly predicted a positive class (1) when the actual class was negative (0)
- True Negative (TN): the number of times the model correctly predicted a negative class (0) when the actual class was negative (0)
- False Negative (FN): the number of times the model incorrectly predicted a negative class (0) when the actual class was positive (1)

Further, the formulas are:

- Accuracy = (TP + TN) / (TP + TN + FP + FN)
- Specificity = TN / (TN + FP)
- Sensitivity (Recall or True Positive Rate) = TP / (TP + FN)
- Precision = TP / (TP + FP)
- F1 Score = 2 * Precision * Recall / (Precision + Recall)

ROC (Receiver Operating Characteristic) Curve:

- Definition: A curve that plots the True Positive Rate (TPR) against the False Positive Rate (FPR) for different classification thresholds.
- Meaning: The AUC (Area Under the Curve) of the ROC curve is often used as a performance metric for binary classification models.

For Neural Network Model here:

- TP = 34.70
- FP = 5.30
- TN = 95.00
- FN = 65.30
- Accuracy = (34.70 + 95.00) / (34.70 + 5.30 + 95.00 + 65.30) = 0.6545 (65.45%)
- Specificity = 95.00 / (95.00 + 5.30) = 0.9476 (94.76%)
- Sensitivity (or Recall) = 34.70 / (34.70 + 65.30) = 0.3467 (34.67%)
- Precision = 34.70 / (34.70 + 5.30) = 0.8675 (86.75%)
- F1 Score = 2 * 0.8675 * 0.3467 / (0.8675 + 0.3467) = 0.4967 (49.67%)

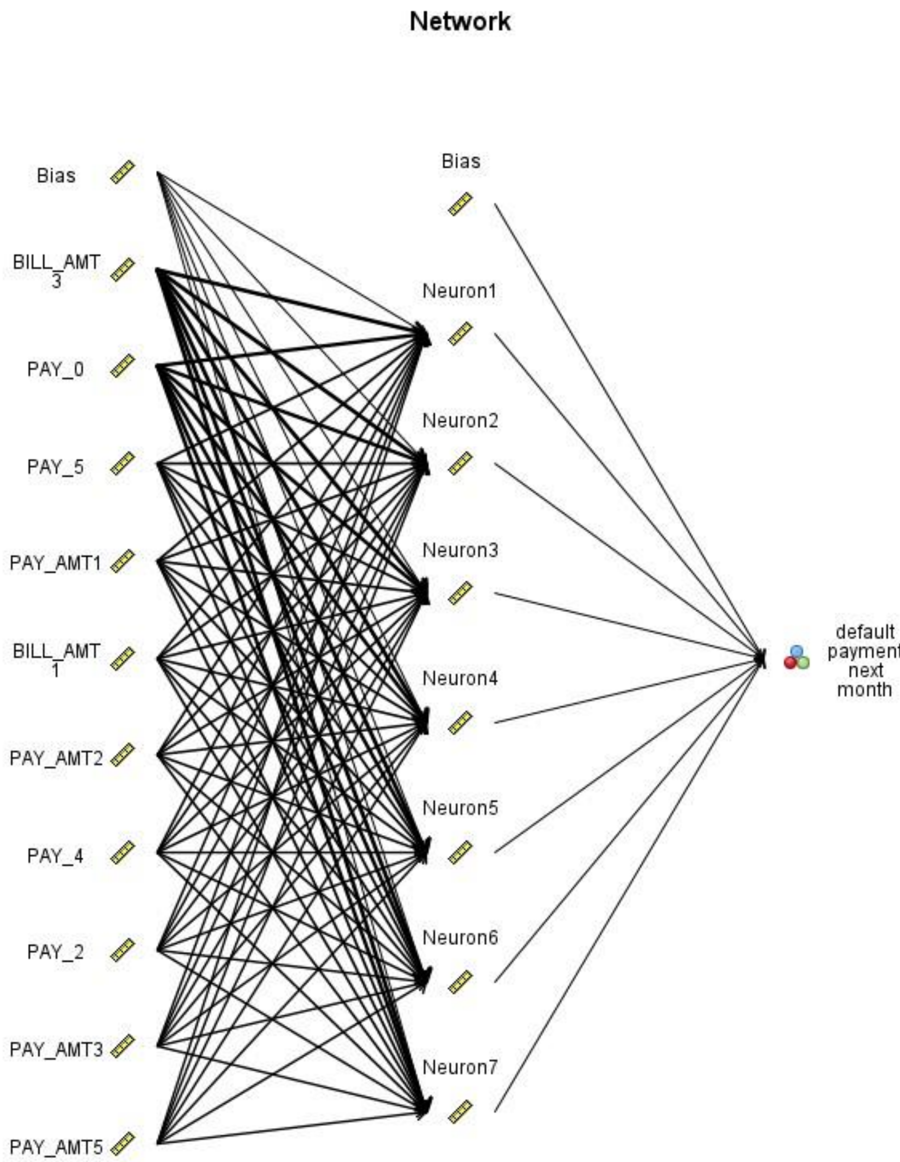


Figure: Neural Network Model

Support Vector Machine

Results for output field default payment next month

Individual Models

Comparing $S-default payment next month with default payment next month

'Partition'	1_Training		2_Testing	
Correct	17,052	81.3%	7,245	80.26%
Wrong	3,921	18.7%	1,782	19.74%
Total	20,973		9,027	

Coincidence Matrix for $S-default payment next month (rows show actuals)

'Partition' = 1_Training	0.000000	1.000000
0.000000	15,939	509
1.000000	3,412	1,113

'Partition' = 2_Testing	0.000000	1.000000
0.000000	6,719	197
1.000000	1,585	526

Performance Evaluation

'Partition' = 1_Training	
0.000000	0.049
1.000000	1.157

'Partition' = 2_Testing	
0.000000	0.055
1.000000	1.135

Evaluation Metrics

'Partition'	1_Training		2_Testing	
Model	AUC	Gini	AUC	Gini
$S-default payment next month	0.723	0.446	0.718	0.435

Figure: Detailed Analysis for SVM Model

CHAID Decision Tree

Comparing $R-default payment next month with default payment next month

'Partition'	1_Training		2_Testing	
Correct	17,303	82.5%	7,319	81.08%
Wrong	3,670	17.5%	1,708	18.92%
Total	20,973		9,027	

Coincidence Matrix for $R-default payment next month (rows show actuals)

'Partition' = 1_Training	No	Yes
No	15,666	782
Yes	2,888	1,637

'Partition' = 2_Testing	No	Yes
No	6,570	346
Yes	1,362	749

Performance Evaluation

'Partition' = 1_Training	
No	0.074
Yes	1.143

'Partition' = 2_Testing	
No	0.078
Yes	1.073

Figure: Detailed Analysis for CHAID Decision Tree

CHAID Decision tree tells us which variables are most important in predicting whether a customer will default on their credit card payments. Here, Payment history is the most important factor.

Pay0 is whether the customer made their most recent credit card payment on time or not.

Pay4 whether the customer made their fourth most recent credit card payment on time or not.

Pay_amt_1 refers to amount of customer's most recent credit card payment.

Bill_amt1 referets to amount of customer's most recent credit card bill. Customers who have larger bills are more likely to default.

Definition: In this example, we're using a CHAID modeling node. CHAID, or Chi-squared Automatic Interaction Detection, is a classification method that builds decision trees by using a particular type of statistics known as chi-square statistics to work out the best places to make the splits in the decision tree.

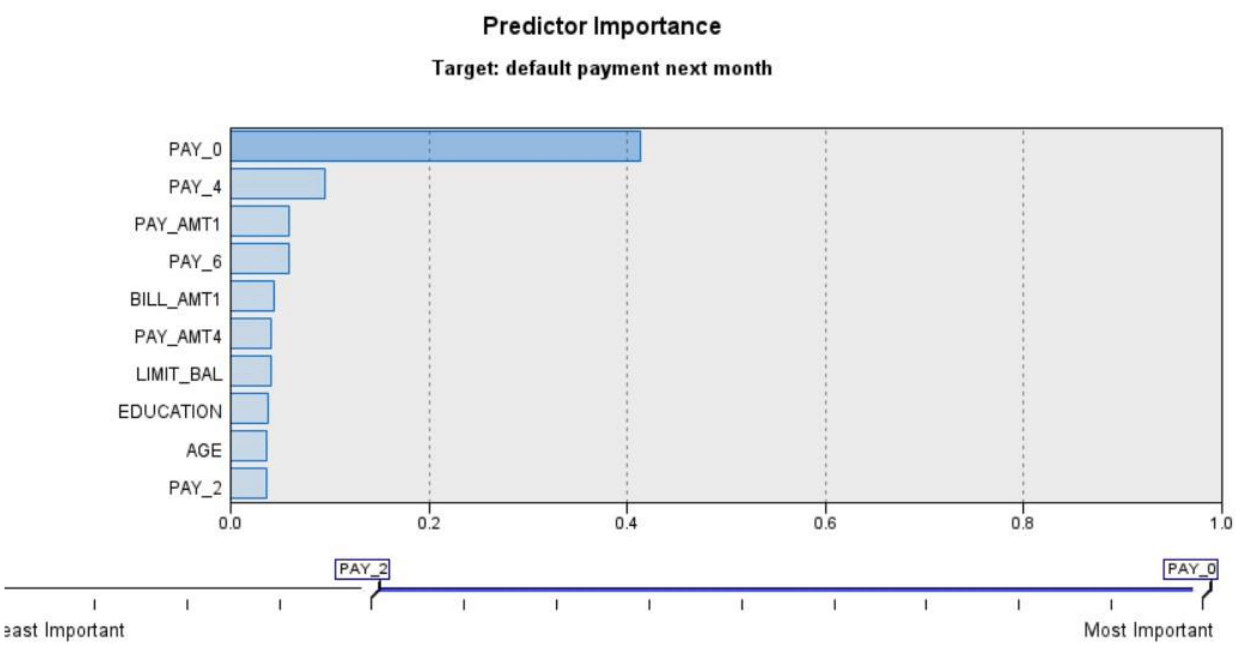


Figure: Predictor Importance for CHAID Decision Tree

By convention, we place the predictor (independent) variables in the rows and the target (dependent) variable in the columns. So, in the matrix node, Target variable is default payment next month and Predictor variable is $R- default payment next month.

```
PAY_0 <= -2 [ Mode: No ]
    PAY_AMT1 <= 0 [ Mode: No ] ⇨ No
    PAY_AMT1 > 0 and PAY_AMT1 <= 4,300 [ Mode: No ] ⇨ No
    PAY_AMT1 > 4,300 [ Mode: No ] ⇨ No
PAY_0 > -2 and PAY_0 <= -1 [ Mode: No ]
    PAY_AMT2 <= 2,016 [ Mode: No ]
        PAY_2 <= 0 [ Mode: No ]
            PAY_AMT4 <= 2,123 [ Mode: No ]
                LIMIT_BAL <= 140,000 [ Mode: No ] ⇨ No
                LIMIT_BAL > 140,000 [ Mode: No ] ⇨ No
            PAY_AMT4 > 2,123 [ Mode: No ] ⇨ No
        PAY_2 > 0 [ Mode: No ] ⇨ No
    PAY_AMT2 > 2,016 and PAY_AMT2 <= 6,024 [ Mode: No ] ⇨ No
    PAY_AMT2 > 6,024 [ Mode: No ]
        BILL_AMT2 <= 5,399 [ Mode: No ] ⇨ No
        BILL_AMT2 > 5,399 [ Mode: No ] ⇨ No
PAY_0 > -1 and PAY_0 <= 0 [ Mode: No ]
    PAY_4 <= 0 [ Mode: No ]
        LIMIT_BAL <= 20,000 [ Mode: No ]
            BILL_AMT1 <= 13,560 [ Mode: No ] ⇨ No
            BILL_AMT1 > 13,560 [ Mode: No ] ⇨ No
        LIMIT_BAL > 20,000 and LIMIT_BAL <= 70,000 [ Mode: No ]
            EDUCATION in [ "OTHERS" "GRADUATE SCHOOL" ] [ Mode: No ] ⇨ No
            EDUCATION in [ "UNIVERSITY" "HIGH SCHOOL" ] [ Mode: No ] ⇨ No
        LIMIT_BAL > 70,000 and LIMIT_BAL <= 140,000 [ Mode: No ]
            BILL_AMT3 <= 20,298 [ Mode: No ] ⇨ No
            BILL_AMT3 > 20,298 and BILL_AMT3 <= 76,755 [ Mode: No ] ⇨ No
            BILL_AMT3 > 76,755 [ Mode: No ]
                SEX in [ "MALE" ] [ Mode: No ] ⇨ No
                SEX in [ "FEMALE" ] [ Mode: No ] ⇨ No
        LIMIT_BAL > 140,000 [ Mode: No ]
```

```
        LIMIT_BAL > 140,000 [ Mode: No ]
            BILL_AMT1 <= 13,560 [ Mode: No ] ⇨ No
            BILL_AMT1 > 13,560 and BILL_AMT1 <= 142,811 [ Mode: No ] ⇨ No
            BILL_AMT1 > 142,811 [ Mode: No ] ⇨ No
    PAY_4 > 0 [ Mode: No ]
        PAY_AMT2 <= 2,016 [ Mode: No ] ⇨ No
        PAY_AMT2 > 2,016 [ Mode: No ] ⇨ No
PAY_0 > 0 and PAY_0 <= 1 [ Mode: No ]
    PAY_4 <= -2 [ Mode: No ] ⇨ No
    PAY_4 > -2 and PAY_4 <= -1 [ Mode: No ] ⇨ No
    PAY_4 > -1 and PAY_4 <= 0 [ Mode: No ]
        PAY_AMT4 <= 2,123 [ Mode: No ]
            PAY_AMT1 <= 0 [ Mode: No ] ⇨ No
            PAY_AMT1 > 0 [ Mode: No ] ⇨ No
        PAY_AMT4 > 2,123 [ Mode: No ] ⇨ No
    PAY_4 > 0 [ Mode: No ]
        SEX in [ "MALE" ] [ Mode: Yes ] ⇨ Yes
        SEX in [ "FEMALE" ] [ Mode: No ] ⇨ No
PAY_0 > 1 [ Mode: Yes ]
    PAY_6 <= -1 [ Mode: Yes ] ⇨ Yes
    PAY_6 > -1 and PAY_6 <= 0 [ Mode: Yes ]
        EDUCATION in [ "OTHERS" "UNIVERSITY" "HIGH SCHOOL" ] [ Mode: Yes ] ⇨ Yes
        EDUCATION in [ "GRADUATE SCHOOL" ] [ Mode: Yes ] ⇨ Yes
    PAY_6 > 0 [ Mode: Yes ]
        AGE <= 39 [ Mode: Yes ] ⇨ Yes
        AGE > 39 [ Mode: Yes ] ⇨ Yes
```

Figure: Tree and Nodes for the CHAID Model, helpful to learn about the splits, nodes represent features and each branch represents a decision based on that feature

Quest Decision Tree

Comparing $R1-default payment next month with default payment next month

'Partition'	1_Training		2_Testing	
Correct	17,287	82.43%	7,301	80.88%
Wrong	3,686	17.57%	1,726	19.12%
Total	20,973		9,027	

Coincidence Matrix for $R1-default payment next month (rows show actuals)

'Partition' = 1_Training	No	Yes
No	15,786	662
Yes	3,024	1,501

'Partition' = 2_Testing	No	Yes
No	6,625	291
Yes	1,435	676

Performance Evaluation

'Partition' = 1_Training	
No	0.068
Yes	1.168

'Partition' = 2_Testing	
No	0.07
Yes	1.095

Figure: Detailed Analysis for QUEST Decision Tree

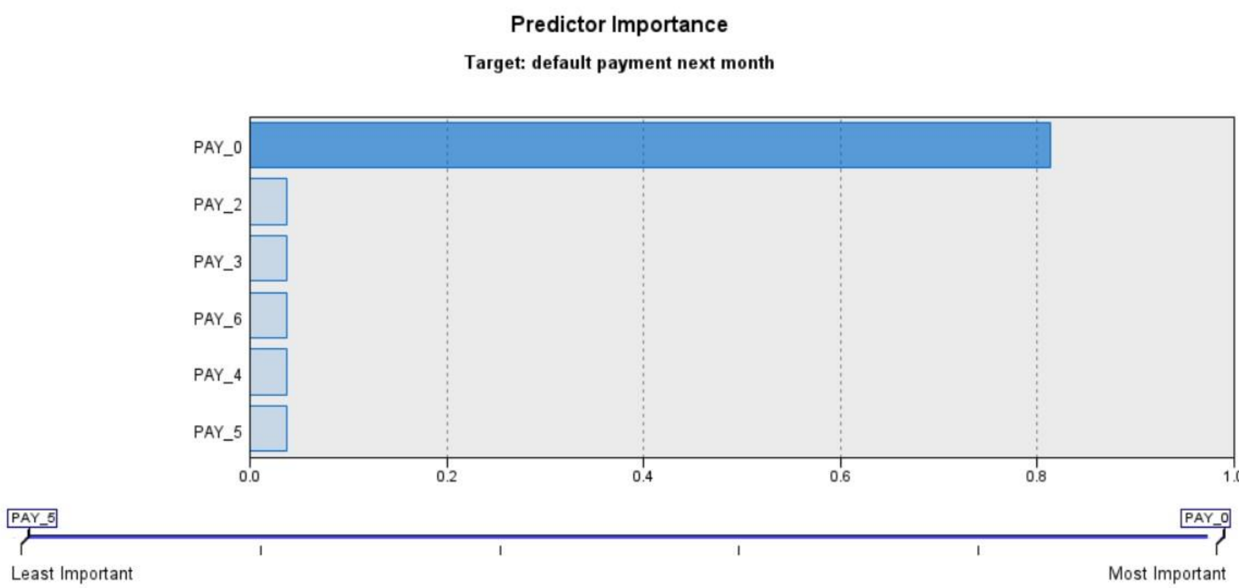


Figure: Predictor Importance for QUEST Decision Tree

CRT Decision Tree

Comparing $S-default payment next month with default payment next month

'Partition'	1_Training		2_Testing	
Correct	17,052	81.3%	7,245	80.26%
Wrong	3,921	18.7%	1,782	19.74%
Total	20,973		9,027	

Coincidence Matrix for $S-default payment next month (rows show actuals)

'Partition' = 1_Training	No	Yes
No	15,939	509
Yes	3,412	1,113

'Partition' = 2_Testing	No	Yes
No	6,719	197
Yes	1,585	526

Performance Evaluation

'Partition' = 1_Training	
No	0.049
Yes	1.157

'Partition' = 2_Testing	
No	0.055
Yes	1.135

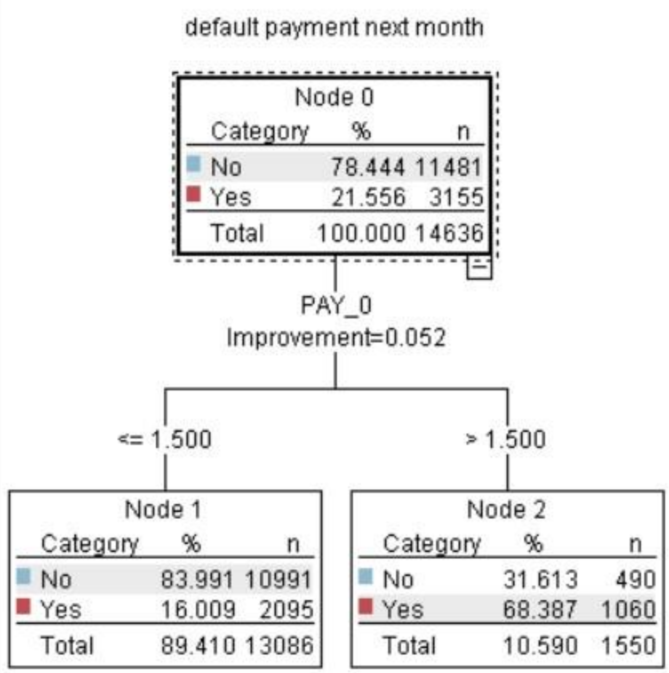


Figure: Nodes represented for CRT Model

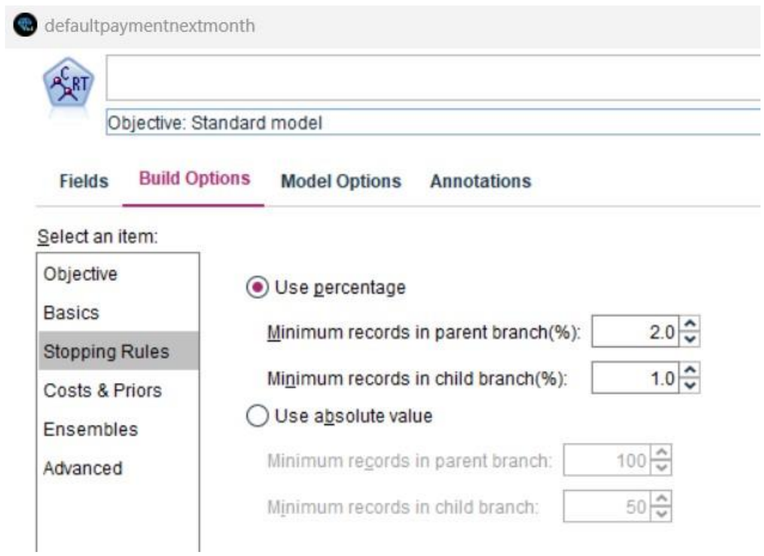


Figure: Stopping rule for CRT

C5.0

Results for output field default payment next month

Individual Models

Comparing $C-default payment next month with default payment next month

'Partition'	1_Training		2_Testing	
Correct	17,337	82.66%	7,318	81.07%
Wrong	3,636	17.34%	1,709	18.93%
Total	20,973		9,027	

Coincidence Matrix for $C-default payment next month (rows show actuals)

'Partition' = 1_Training	0.000000	1.000000
0.000000	15,682	766
1.000000	2,870	1,655
'Partition' = 2_Testing	**0.000000**	**1.000000**
0.000000	6,580	336
1.000000	1,373	738

Performance Evaluation

'Partition' = 1_Training	
0.000000	0.075
1.000000	1.153
'Partition' = 2_Testing	
0.000000	0.077
1.000000	1.078

Evaluation Metrics

'Partition'	1_Training		2_Testing	
Model	AUC	Gini	AUC	Gini
$C-default payment next month	0.696	0.392	0.69	0.38

Figure: Detailed Analysis for C 5.0 Decision tree

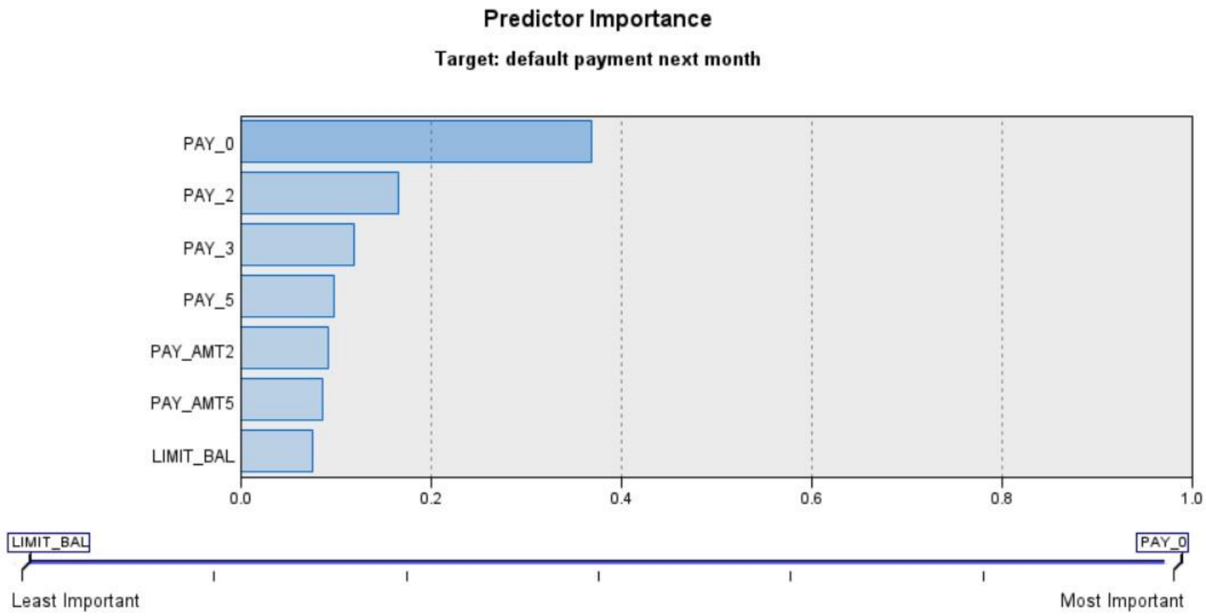


Figure: Predictor Importance for C 5.0

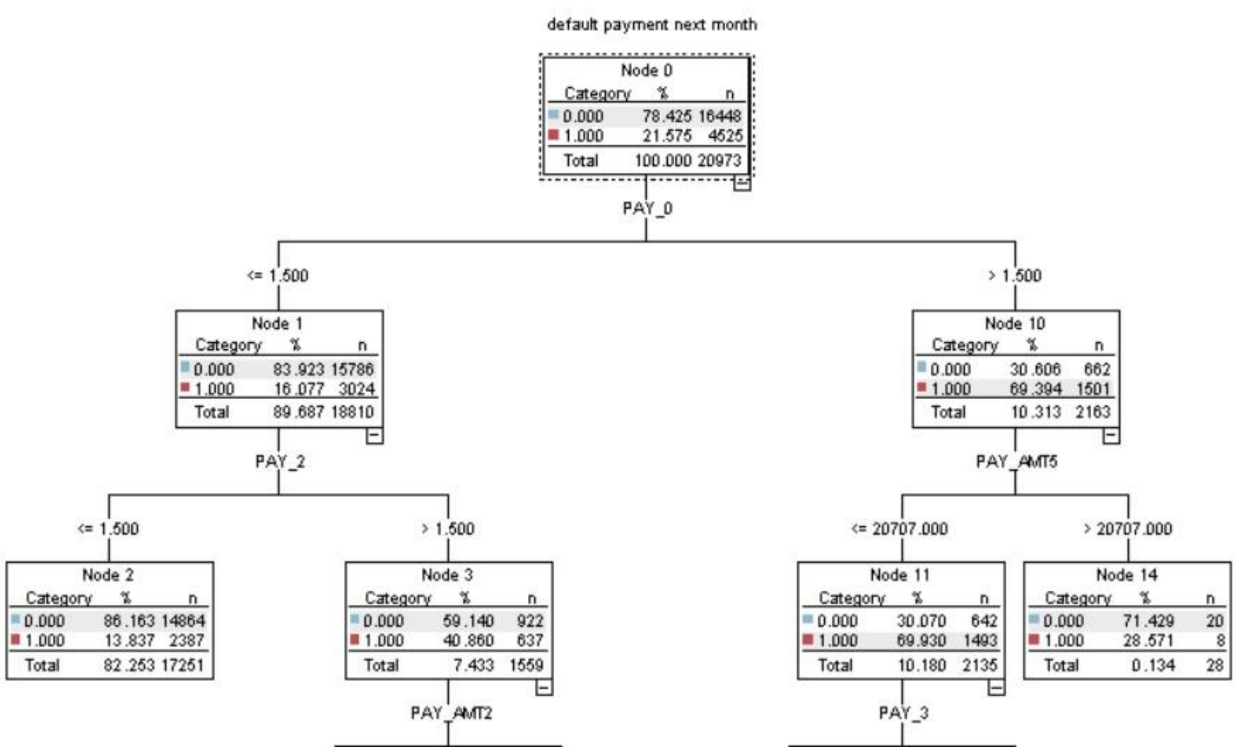


Figure: Decision Tree Part I for C 5.0 algorithm

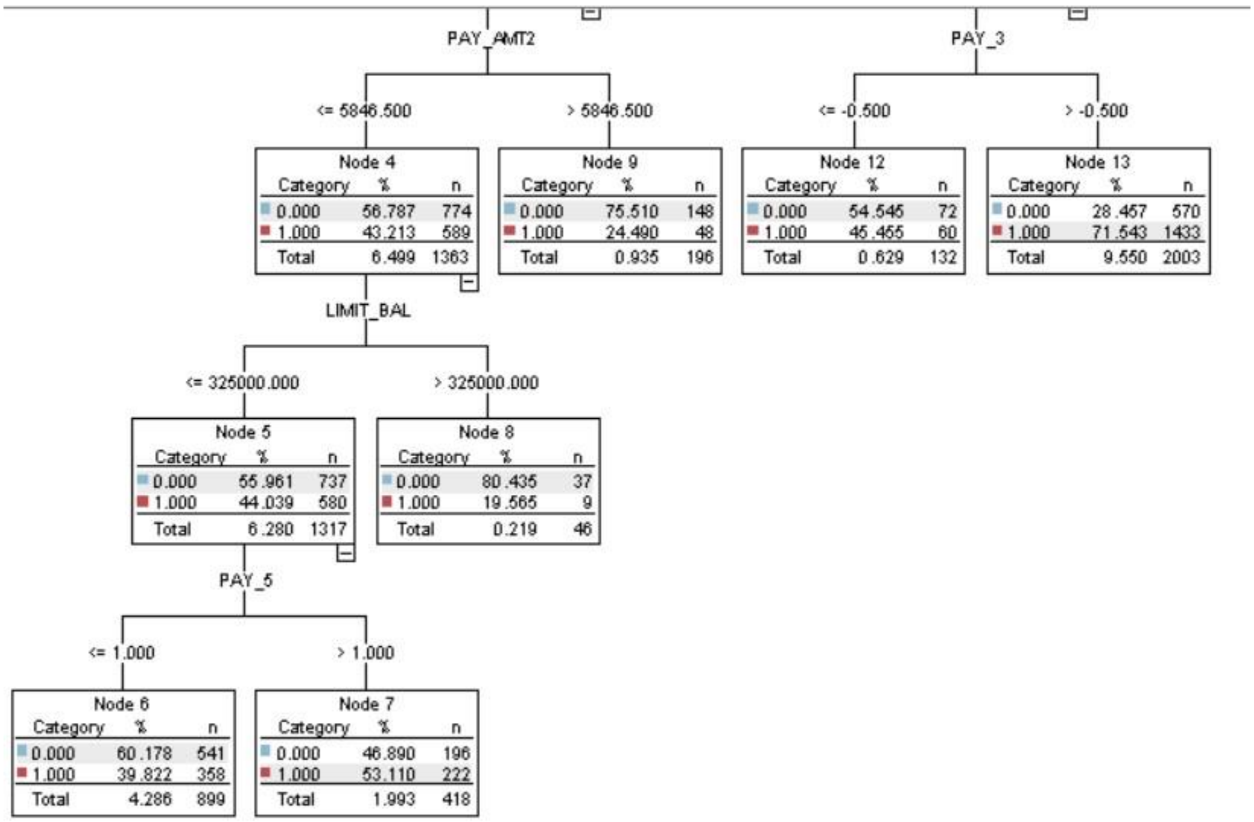


Figure: Decision Tree Part I for C 5.0 algorithm

The figure continues in 2 parts and are connected.

Figure: Detailed analysis after selection of Coincidence matrix, performance evaluation and evaluation metrics are selected (AUC, Gini). It helps us know if we are going in the right direction.

We can say that the C5.0 algorithm correctly classified 17337 cases in Training and 7,318 cases in Testing set. It has a higher true positive rate than false positive rate, good predictive power.

The AUC Score tells us the overall performance of the classifier by calculating the area under the area under the receiver operating characteristic (ROC) curve. It ranges from 0 to 1, where 1 is perfect and 0.5 is almost random. The ROC Curve plots the True positive rate (sensitivity) against the false positive rate(1-Specificity) for different threshold values of classifier.

Gini Score is a metric that measures the inequality among values of a frequency distribution. Also, It can be said that it measures the probability for a random instance being misclassified when chosen randomly. The lower the Gini Index, the better and the lower the chance of misclassification.

Finally, when trying to compare different models, we put the accuracy score on the testing set.

Model Classifier	**Accuracy**
KNN	78.8
Neural Network	80.66
SVM	80.26
CHAID	**81.08**
QUEST	80.88
CRT	80.26
C 5.0	81.07

Auto Classifier

We put all the 18 models in Auto Classifier and it classified based on the following 18 models. The next table shows all the data of 10 models. You can see the graphs, model, build time, number of fields used, overall accuracy, area under the curve, accumulated accuracy and accumulated AUC.

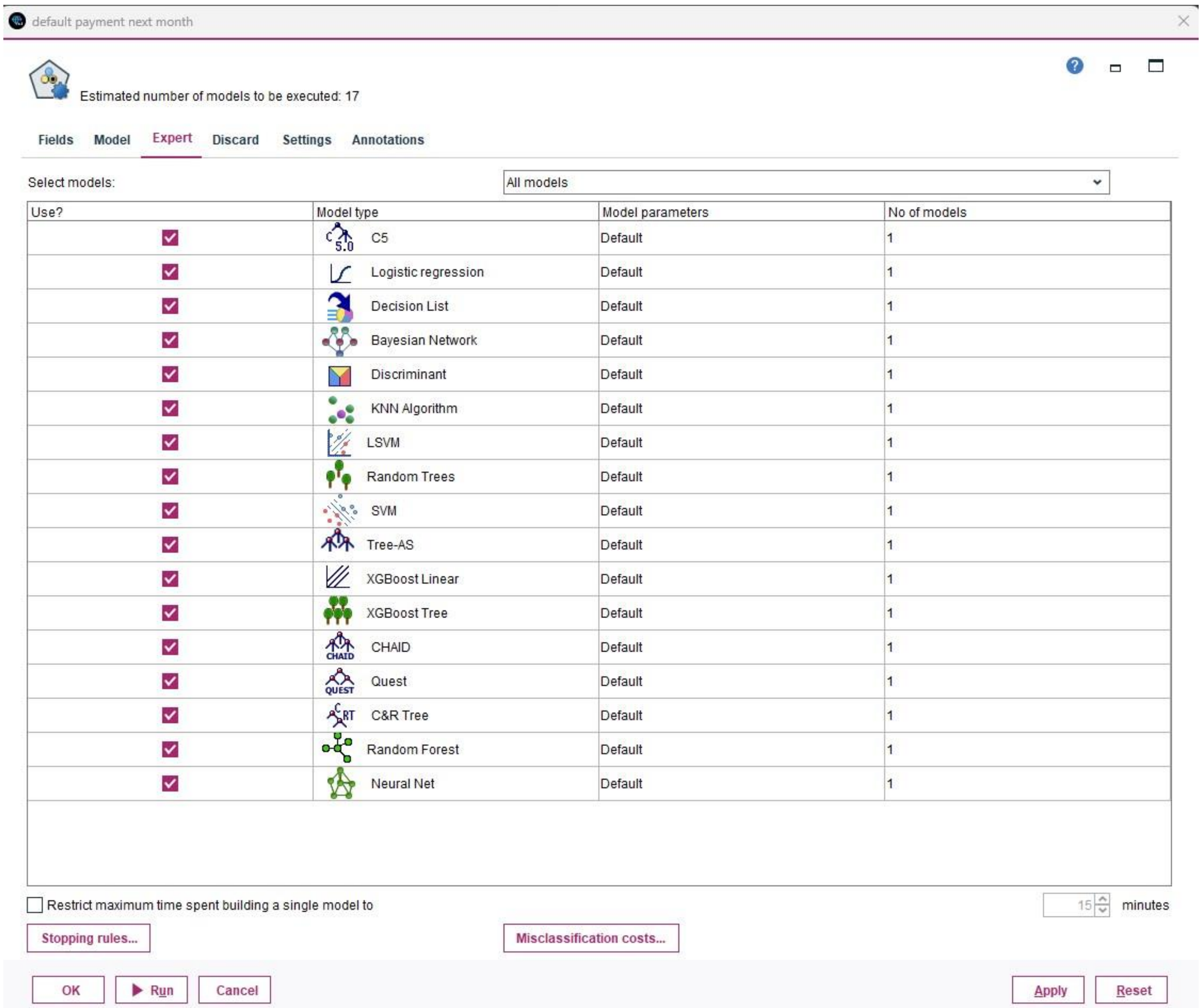


Figure: You can see all the 17 models that are run from which 10 models are selected and kept.

Sort by: Use ○ Ascending ◉ Descending ✕ Delete Unused Models View: Testing set

Use?	Graph	Model	Build Time (mins)	Max Profit	Max Profit Occurs in (%)	Lift{Top 30%}	No. Fields Used	Overall Accuracy (%)	Area Under Curve	Accumulated Accuracy (%)	Accumulated AUC
☑		CHAID 1	5	4,240.0	10	2.098	14	82.501	0.777	82.501	0.777
☑		Neural Net 1	5	3,780.0	11	2.013	23	82.001	0.749	82.001	0.749
☑		Tree-AS 1	5	3277.790	14	2.112	21	81.562	0.779	81.562	0.779
☑		Logistic regression 1	4	3,915.0	10	1.973	23	81.510	0.727	81.510	0.727
☑		SVM 1	5	3,520.0	12	2.039	23	81.305	0.723	81.305	0.723
☑		Bayesian Network 1	4	2,993.571	10	2.048	23	81.276	0.756	81.276	0.756
☑		LSVM 1	5	3,810.0	11	1.965	23	80.699	0.724	80.699	0.724
☑		XGBoost Tree 1	5	660.0	11	2.081	23	78.425	0.779	78.425	0.779
☑		Decision List 1	4	3816.768	10	2.010	7	76.236	0.722	76.236	0.722
☑		Discriminant 1	4	3,780.0	11	1.960	20	73.180	0.717	73.180	0.717

Figure: It shows testing results, sorted in descending order based on Accuracy.

Sort by: Area under curve ○ Ascending ◉ Descending ✕ Delete Unused Models View: Testing set

Use?	Graph	Model	Accumulated Accuracy (%)	Accumulated AUC	No. Fields Used
☑		CHAID 1	82.501	0.777	14
☑		Neural Net 1	82.001	0.749	23
☑		Tree-AS 1	81.562	0.779	21
☑		Logistic regression 1	81.510	0.727	23
☑		SVM 1	81.305	0.723	23
☑		Bayesian Network 1	81.276	0.756	23
☑		LSVM 1	80.699	0.724	23
☑		XGBoost Tree 1	78.425	0.779	23
☑		Decision List 1	76.236	0.722	7
☑		Discriminant 1	73.180	0.717	20

When you run the Auto Classifier tool and specify a filter value of 10, it will create and evaluate all 17 models and rank them, but only the top 10 models that meet your filter criteria will be displayed in the output. The filter criteria can be based on various metrics such as accuracy, sensitivity, specificity, and kappa coefficient, among others.

Accuracy on Testing and Testing Set and comments:

Accuracy measures how often the model correctly predicts the class of an instance, while AUC measures the ability of the model to distinguish between the classes.

In the given table, all models have a relatively high overall accuracy percentage (>73%), so we have good performance.

The models compared are CHAID, Neural Net, Tree-AS, Logistic Regression, SVM, Bayesian Network, LSVM, XGBoost, Decision List and Discriminant, respectively.

Model	Accuracy	AUC	Max Profit	Max Profit Occurs (%)	Lift (Top 30%)
CHAID	82.5	0.78	4240	10	2.1
Neural Net	82	0.75	3780	11	2.01
Tree-AS	81.56	0.78	3277.79	14	2.11
Logistic Regression	81.51	0.73	3915	10	1.97
SVM	81.3	0.72	3520	12	2.04
Bayesian Network	81.28	0.76	2993.57	10	2.05
LSVM	80.7	0.72	3810	11	1.96
XGBoost	78.42	0.78	660	11	2.08
Decision List	76.24	0.72	3816.77	10	2.01
Discriminant Analysis	73.18	0.72	3780	11	1.96

The Key Takeaway:

Accuracy:

- CHAID, Neural Net, Tree-AS and Bayesian Network perform good with accuracy range from 81.56% to 82.50%. Prefer these models if primary goal is to accurately predict outcome.
- XGBoost have an accuracy score of 76.24% but has a high AUC score of 0.72, meaning it's good for ranking predicted outcomes. Can be used if that is the goal.

AUC:

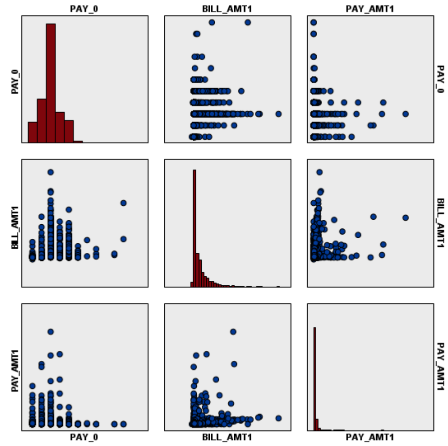
- CHAID, Neural Net and Beyesian Network has high AUC Scores with range from 0.77 to 0.78. If goal it to max out the Area under the ROC curve, and focus on distinguishing between positive and negative outcome

Profit

- CHAID Model has the highest profit of 4240, at rate of 10%. Next best is Neural Net.

Lift

- CHAID, Tree-AS, Bayesian Network have high scores.



The CHAID 1 model has the highest overall accuracy (82.50131%) and the highest area under the curve (0.7773336). It also has the highest accumulated accuracy (82.50131%) and accumulated AUC (0.7773336). Therefore, it is the best machine learning model for the UCI Default on credit card dataset based on the evaluation metrics considered, followed by the neural net and tree models.

The Partition has been done on 70 and 30 ratios.

Convention of Prefix (Excerpt from book IBM SPSS Modeler 18.3):

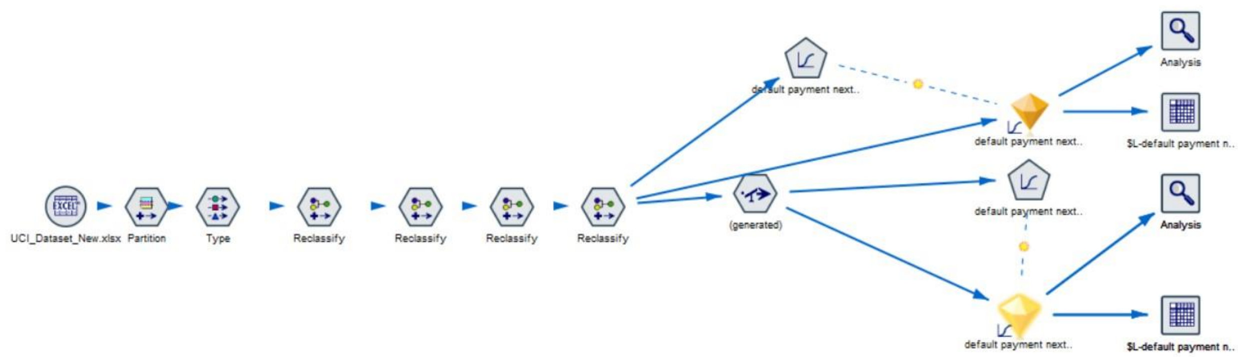
For association rule sets, the prefixes are $A- for the prediction field and $AC- for the confidence field. For C5.0 rule sets, the prefixes are $C- for the prediction field and $CC for the confidence field. For C&R Tree rule sets, the prefixes are $R- for the prediction field and $RC- for the confidence field. In a stream with multiple Rule Set nuggets in a series predicting the same output field(s), the new field names will include numbers in the prefix to distinguish them from each other. The first association Rule Set nugget in the stream will use the usual names, the second node will use names starting with $A1- and $AC1-, the third node will use names starting with $A2- and $AC2-, and so on.

Predictor Importance (Page 128 IBM SPSS Modeler 18.3 Modeling Nodes):

Typically, you will want to focus your modeling efforts on the predictor fields that matter most and consider dropping or ignoring those that matter least. The predictor importance chart helps you do this by indicating the relative importance of each predictor in estimating the model. Since the values are relative, the sum of the values for all predictors on the display is 1.0. Predictor importance does not relate to model accuracy. It just relates to the importance of each predictor in making a prediction, not whether or not the prediction is accurate.

Neural Net (From the book)
Predictor importance. Connecting lines in the diagram are weighted based on predictor importance, with greater line width corresponding to greater importance. There is a Predictor Importance slider in the toolbar that controls which predictors are shown in the network diagram. This does not change the model, but simply allows you to focus on the most important predictors.

Logistic Regression



Before Balancing

Results for output field default payment next month

Individual Models

Comparing $L-default payment next month with default payment next month

'Partition'	1_Training		2_Testing	
Correct	16,443	81.59%	7,862	79.85%
Wrong	3,711	18.41%	1,984	20.15%
Total	20,154		9,846	

Coincidence Matrix for $L-default payment next month (rows show actuals)

'Partition' = 1_Training	No	Yes
No	15,425	402
Yes	3,309	1,018

'Partition' = 2_Testing	No	Yes
No	7,329	208
Yes	1,776	533

Performance Evaluation

'Partition' = 1_Training	
No	0.047
Yes	1.206

'Partition' = 2_Testing	
No	0.05
Yes	1.121

Evaluation Metrics

'Partition'	1_Training		2_Testing	
Model	AUC	Gini	AUC	Gini
$L-default payment next month	0.726	0.452	0.723	0.447

Figure: Detailed Analysis for Logistic Regression before balancing default payment next month

$L-default payment next month	No	Yes
No	22754	5085
Yes	610	1551

Cells contain: cross-tabulation of fields (including missing values)

Chi-square = 3,332.656, df = 1, probability = 0

After Balancing

Results for output field default payment next month

Individual Models

Comparing $L-default payment next month with default payment next month

'Partition'	1_Training		2_Testing	
Correct	16,443	81.59%	7,862	79.85%
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Cells contain: cross-tabulation of fields (including missing values)

Chi-square = 3,332.656, df = 1, probability = 0

Summary for Logistic Regression Module:

The same before and after balancing which validates our hypothesis.

Case Processing Summary

		N	Marginal Percentage
default payment next month	No	15827	78.5%
	Yes	4327	21.5%
SEX	Female	12181	60.4%
	Male	7973	39.6%
EDUCATION	Graduate School	7104	35.2%
	High School	3299	16.4%
	Others	315	1.6%
	University	9436	46.8%
MARRIAGE	Married	9168	45.5%
	Others	212	1.1%
	Single	10729	53.2%
	Unknown	45	0.2%
Valid		20154	100.0%
Missing		0	
Total		20154	
Subpopulation		20124[a]	

a. The dependent variable has only one value observed in 20111 (99.9%) subpopulations.

Figure: We go to Advance section of model to look further

Parameter Estimates

default payment next month[a]		B	Std. Error	Wald	df	Sig.	Exp(B)	95% Confidence Interval for Exp (B) Lower Bound	Upper Bound
Yes	Intercept	-2.780	.675	16.978	1	<.001			
	LIMIT_BAL	.000	.000	10.113	1	.001	1.000	1.000	1.000
	AGE	.005	.002	5.559	1	.018	1.005	1.001	1.010
	PAY_0	.588	.022	725.393	1	<.001	1.800	1.724	1.878
	PAY_2	.081	.025	10.728	1	.001	1.085	1.033	1.139
	PAY_3	.076	.028	7.581	1	.006	1.079	1.022	1.139
	PAY_4	.031	.031	.997	1	.318	1.031	.971	1.095
	PAY_5	.065	.033	3.816	1	.051	1.067	1.000	1.139
	PAY_6	-.038	.028	1.938	1	.164	.962	.912	1.016
	BILL_AMT1	.000	.000	13.327	1	<.001	1.000	1.000	1.000
	BILL_AMT2	.000	.000	2.275	1	.131	1.000	1.000	1.000
	BILL_AMT3	.000	.000	.441	1	.506	1.000	1.000	1.000
	BILL_AMT4	.000	.000	.004	1	.953	1.000	1.000	1.000
	BILL_AMT5	.000	.000	.022	1	.883	1.000	1.000	1.000
	BILL_AMT6	.000	.000	.012	1	.912	1.000	1.000	1.000
	PAY_AMT1	.000	.000	27.160	1	<.001	1.000	1.000	1.000
	PAY_AMT2	.000	.000	7.631	1	.006	1.000	1.000	1.000
	PAY_AMT3	.000	.000	1.054	1	.305	1.000	1.000	1.000
	PAY_AMT4	.000	.000	1.284	1	.257	1.000	1.000	1.000
	PAY_AMT5	.000	.000	4.761	1	.029	1.000	1.000	1.000
	PAY_AMT6	.000	.000	1.122	1	.289	1.000	1.000	1.000
	[SEX=Female]	-.114	.038	9.121	1	.003	.892	.828	.961
	[SEX=Male]	0[b]	.	.	0	.	.	.	.
	[EDUCATION=Graduate School]	.111	.044	6.332	1	.012	1.117	1.025	1.217
	[EDUCATION=High School]	.035	.052	.456	1	.500	1.036	.935	1.148
	[EDUCATION=Others]	-1.223	.246	24.745	1	<.001	.294	.182	.477
	[EDUCATION=University]	0[b]	.	.	0	.	.	.	.
	[MARRIAGE=Married]	1.638	.669	5.998	1	.014	5.146	1.387	19.093
	[MARRIAGE=Others]	1.513	.690	4.811	1	.028	4.538	1.175	17.533
	[MARRIAGE=Single]	1.448	.669	4.682	1	.030	4.253	1.146	15.786
	[MARRIAGE=Unknown]	0[b]	.	.	0	.	.	.	.

a. The reference category is: No.

b. This parameter is set to zero because it is redundant.

Model Fitting Information

Model	Model Fitting Criteria: -2 Log Likelihood	Likelihood Ratio Tests: Chi-Square	df	Sig.
Intercept Only	20945.902			
Final	18369.331	2576.571	27	.000

Pseudo R-Square

Cox and Snell	.120
Nagelkerke	.186
McFadden	.123

Figure: We check the warning to know if there is a need to correct, inspecting the beta, the p value and the nodes

Section 4: Evaluation

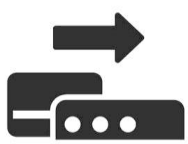
Call to Action

Summary

After performing the data mining analysis on the dataset, we were able to achieve our business goals by identifying potential customers who are at higher risk of defaulting on their credit card payments.

We find a positive correlation between Age and credit limit. So, as age increases people have access to higher credit limits. Further, we did customer segmentation and have a Marketing campaign with Age and Credit Limit.

Further, in classification we find our models have accuracy of more than 80% indicating that our classification models performed well. Also, the ability to predict who's not going to default up to 95% helps you know which customers you don't have to worry about.



The first business goal was to reduce credit card default rates by identifying customers who are at higher risk of defaulting. After analysis, we identified that 22% of customers defaulted on their credit card payments. We also found that customers who are younger, have lower education levels, and are unmarried are more likely to default. **By targeting these high-risk customers with advice on credit and better payment plans, we will be able to reduce the default rate by good %.**

The second business goal was to increase customer satisfaction and retention by understanding their financial behavior and needs. We were able to identify different segments of customers based on their financial behavior and credit use. Further, we will be able to **offer customized credit card products and services, resulting in increased customer satisfaction and retention.**

The third business goal was to develop targeted marketing campaigns to attract new customers and retain existing ones based on their credit card usage. After analysis, we are able to identify patterns and trends in credit card use among different customer segments. **By developing marketing campaigns targeted based on those patterns and trends, we will be able to increase the number of new customers and retain existing ones.**

Appendix

Research Papers

Yeh, I. C., & Lien, C. H. (2009). The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients. Expert Systems with Applications, 36(2), 2473-2480. https://doi.org/10.1016/j.eswa.2008.01.048

Logistic Regression: Hosmer, D. W., Jr., Lemeshow, S., & Sturdivant, R. X. (2013). Applied logistic regression (Vol. 398). John Wiley & Sons.

Decision Trees: Breiman, L., Friedman, J., Stone, C. J., & Olshen, R. A. (1984). Classification and regression trees. Chapman and Hall/CRC.

Random Forest: Breiman, L. (2001). Random forests. Machine learning, 45(1), 5-32.

Gradient Boosting: Friedman, J. H. (2001). Greedy function approximation: a gradient boosting machine. Annals of statistics, 1189-1232.

Witten, I. H., & Frank, E. (2005). Data mining: Practical machine learning tools and techniques (2nd ed.). Morgan Kaufmann Publishers.

Hastie, T., Tibshirani, R., & Friedman, J. (2009). The elements of statistical learning: Data mining, inference, and prediction (2nd ed.). Springer.

Li, X., & Han, J. (2010). Data mining techniques and applications: An introduction. CRC Press.

Hawkins (1980), "The Detection of Outliers"

Crook, J., Edelman, D. B., & Thomas, L. C. (2007). Recent developments in consumer credit modeling. Journal of Banking and Finance, 31(11), 3069-3082. doi: 10.1016/j.jbankfin.2007.06.004

Breiman, L. (2001). Random forests. Machine Learning, 45(1), 5-32. doi: 10.1023/A:1010933404324

Liu, Y., Wang, C., & Wang, S. (2019). The impact of big data and machine learning on credit risk management. Journal of Financial Stability, 43, 45-57. doi: 10.1016/j.jfs.2019.08.005

https://exploringtm1.com/spss-modeler-15-use-auto-classifier-node/

Notes From Book

Book "IBM SPSS Modeler 18.3 Modeling Nodes":

All the algorithms and nodes are learned and applied as per convention and terminology mentioned in the book by IBM Corporation. (2017). IBM SPSS Modeler 18.3 Modeling Nodes. IBM Corporation.

Further, if you wish to learn the background of the algorithms used in Data Mining, Descriptive, Diagnostic and Predictive Modelling you can see in the pictures shown.

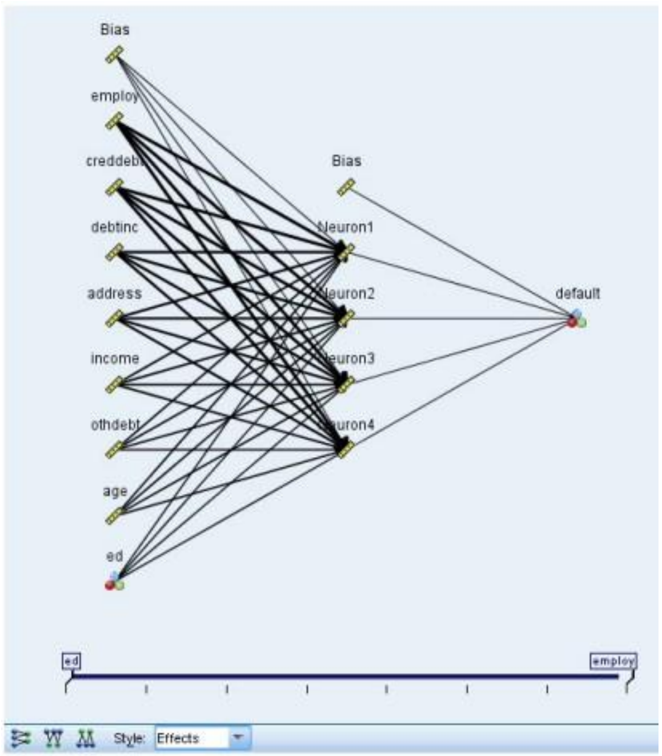


Figure 42. Network view, inputs on the left, effects style

Chapter 5. Automated Modeling Nodes

The automated modeling nodes estimate and compare a number of different modeling methods, enabling you to try out a variety of approaches in a single modeling run. You can select the modeling algorithms to use, and the specific options for each, including combinations that would otherwise be mutually-exclusive. For example, rather than choose between the quick, dynamic, or prune methods for a Neural Net, you can try them all. The node explores every possible combination of options, ranks each candidate model based on the measure you specify, and saves the best for use in scoring or further analysis.

You can choose from three automated modeling nodes, depending on the needs of your analysis:

The Auto Classifier node creates and compares a number of different models for binary outcomes (yes or no, churn or do not churn, and so on), allowing you to choose the best approach for a given analysis. A number of modeling algorithms are supported, making it possible to select the methods you want to use, the specific options for each, and the criteria for comparing the results. The node generates a set of models based on the specified options and ranks the best candidates according to the criteria you specify.

The Auto Numeric node estimates and compares models for continuous numeric range outcomes using a number of different methods. The node works in the same manner as the Auto Classifier node, allowing you to choose the algorithms to use and to experiment with multiple combinations of options in a single modeling pass. Supported algorithms include neural networks, C&R Tree, CHAID, linear regression, generalized linear regression, and support vector machines (SVM). Models can be compared based on correlation, relative error, or number of variables used.

The Auto Cluster node estimates and compares clustering models, which identify groups of records that have similar characteristics. The node works in the same manner as other automated modeling nodes, allowing you to experiment with multiple combinations of options in a single modeling pass. Models can be compared using basic measures with which to attempt to filter and rank the usefulness of the cluster models, and provide a measure based on the importance of particular fields.

The best models are saved in a single composite model nugget, enabling you to browse and compare them, and to choose which models to use in scoring.

- For binary, nominal, and numeric targets only you can select multiple scoring models and combine the scores in a single model ensemble. By combining predictions from multiple models, limitations in individual models may be avoided, often resulting in a higher overall accuracy than can be gained from any one of the models.
- Optionally, you can choose to drill down into the results and generate modeling nodes or model nuggets for any of the individual models you want to use or explore further.

Models and Execution Time

Depending on the dataset and the number of models, automated modeling nodes may take hours or even longer to execute. When selecting options, pay attention to the number of models being produced. When practical, you may want to schedule modeling runs during nights or weekends when system resources are less likely to be in demand.

- If necessary, a Partition or Sample node can be used to reduce the number of records included in the initial training pass. Once you have narrowed the choices to a few candidate models, the full dataset can be restored.

Supported Algorithms

The Support Vector Machine (SVM) node enables you to classify data into one of two groups without overfitting. SVM works well with wide data sets, such as those with a very large number of input fields.

The *k*-Nearest Neighbor (KNN) node associates a new case with the category or value of the *k* objects nearest to it in the predictor space, where *k* is an integer. Similar cases are near each other and dissimilar cases are distant from each other.

Discriminant analysis makes more stringent assumptions than logistic regression but can be a valuable alternative or supplement to a logistic regression analysis when those assumptions are met.

The Bayesian Network node enables you to build a probability model by combining observed and recorded evidence with real-world knowledge to establish the likelihood of occurrences. The node focuses on Tree Augmented Naïve Bayes (TAN) and Markov Blanket networks that are primarily used for classification.

Chapter 5. Automated Modeling Nodes **57**

The Decision List node identifies subgroups, or segments, that show a higher or lower likelihood of a given binary outcome relative to the overall population. For example, you might look for customers who are unlikely to churn or are most likely to respond favorably to a campaign. You can incorporate your business knowledge into the model by adding your own custom segments and previewing alternative models side by side to compare the results. Decision List models consist of a list of rules in which each rule has a condition and an outcome. Rules are applied in order, and the first rule that matches determines the outcome.

Logistic regression is a statistical technique for classifying records based on values of input fields. It is analogous to linear regression but takes a categorical target field instead of a numeric range.

The CHAID node generates decision trees using chi-square statistics to identify optimal splits. Unlike the C&R Tree and QUEST nodes, CHAID can generate nonbinary trees, meaning that some splits have more than two branches. Target and input fields can be numeric range (continuous) or categorical. Exhaustive CHAID is a modification of CHAID that does a more thorough job of examining all possible splits but takes longer to compute.

The QUEST node provides a binary classification method for building decision trees, designed to reduce the processing time required for large C&R Tree analyses while also reducing the tendency found in classification tree methods to favor inputs that allow more splits. Input fields can be numeric ranges (continuous), but the target field must be categorical. All splits are binary.

The Classification and Regression (C&R) Tree node generates a decision tree that allows you to predict or classify future observations. The method uses recursive partitioning to split the training records into segments by minimizing the impurity at each step, where a node in the tree is considered "pure" if 100% of cases in the node fall into a specific category of the target field. Target and input fields can be numeric ranges or categorical (nominal, ordinal, or flags); all splits are binary (only two subgroups).

The C5.0 node builds either a decision tree or a rule set. The model works by splitting the sample based on the field that provides the maximum information gain at each level. The target field must be categorical. Multiple splits into more than two subgroups are allowed.

The Neural Net node uses a simplified model of the way the human brain processes information. It works by simulating a large number of interconnected simple processing units that resemble abstract versions of neurons. Neural networks are powerful general function estimators and require minimal statistical or mathematical knowledge to train or apply.

Linear regression models predict a continuous target based on linear relationships between the target and one or more predictors.

The Linear Support Vector Machine (LSVM) node enables you to classify data into one of two groups without overfitting. LSVM is linear and works well with wide data sets, such as those with a very large number of records.

58 IBM SPSS Modeler 18.3 Modeling Nodes

The Random Trees node is similar to the existing C&RT node; however, the Random Trees node is designed to process big data to create a single tree and displays the resulting model in the output viewer that was added in SPSS Modeler version 17. The Random Trees tree node generates a decision tree that you use to predict or classify future observations. The method uses recursive partitioning to split the training records into segments by minimizing the impurity at each step, where a node in the tree is considered *pure* if 100% of cases in the node fall into a specific category of the target field. Target and input fields can be numeric ranges or categorical (nominal, ordinal, or flags); all splits are binary (only two subgroups).

The Tree-AS node is similar to the existing CHAID node; however, the Tree-AS node is designed to process big data to create a single tree and displays the resulting model in the output viewer that was added in SPSS Modeler version 17. The node generates a decision tree by using chi-square statistics (CHAID) to identify optimal splits. This use of CHAID can generate nonbinary trees, meaning that some splits have more than two branches. Target and input fields can be numeric range (continuous) or categorical. Exhaustive CHAID is a modification of CHAID that does a more thorough job of examining all possible splits but takes longer to compute.

XGBoost Tree© is an advanced implementation of a gradient boosting algorithm with a tree model as the base model. Boosting algorithms iteratively learn weak classifiers and then add them to a final strong classifier. XGBoost Tree is very flexible and provides many parameters that can be overwhelming to most users, so the XGBoost Tree node in SPSS Modeler exposes the core features and commonly used parameters. The node is implemented in Python.

XGBoost© is an advanced implementation of a gradient boosting algorithm. Boosting algorithms iteratively learn weak classifiers and then add them to a final strong classifier. XGBoost is very flexible and provides many parameters that can be overwhelming to most users, so the XGBoost-AS node in SPSS Modeler exposes the core features and commonly used parameters. The XGBoost-AS node is implemented in Spark.

Note: If you select Tree-AS to run on Analytic Server, it will fail to build a model when there is a Partition node upstream. In this case, to make Auto Classifier work with other modeling nodes on Analytic Server, deselect the Tree-AS model type.

Tree-building algorithms

Several algorithms are available for performing classification and segmentation analysis. These algorithms all perform basically the same thing, they examine all of the fields of your dataset to find the one that gives the best classification or prediction by splitting the data into subgroups. The process is applied recursively, splitting subgroups into smaller and smaller units until the tree is finished (as defined by certain stopping criteria). The target and input fields used in tree building can be continuous (numeric range) or categorical, depending on the algorithm used. If a continuous target is used, a regression tree is generated; if a categorical target is used, a classification tree is generated.

The Classification and Regression (C&R) Tree node generates a decision tree that allows you to predict or classify future observations. The method uses recursive partitioning to split the training records into segments by minimizing the impurity at each step, where a node in the tree is considered "pure" if 100% of cases in the node fall into a specific category of the target field. Target and input fields can be numeric ranges or categorical (nominal, ordinal, or flags); all splits are binary (only two subgroups).

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The C5.0 node builds either a decision tree or a rule set. The model works by splitting the sample based on the field that provides the maximum information gain at each level. The target field must be categorical. Multiple splits into more than two subgroups are allowed.

The Tree-AS node is similar to the existing CHAID node; however, the Tree-AS node is designed to process big data to create a single tree and displays the resulting model in the output viewer that was added in SPSS Modeler version 17. The node generates a decision tree by using chi-square statistics (CHAID) to identify optimal splits. This use of CHAID can generate nonbinary trees, meaning that some splits have more than two branches. Target and input fields can be numeric range (continuous) or categorical. Exhaustive CHAID is a modification of CHAID that does a more thorough job of examining all possible splits but takes longer to compute.

The Random Trees node is similar to the existing C&RT node; however, the Random Trees node is designed to process big data to create a single tree and displays the resulting model in the output viewer that was added in SPSS Modeler version 17. The Random Trees tree node generates a decision tree that you use to predict or classify future observations. The method uses recursive partitioning to split the training records into segments by minimizing the impurity at each step, where a node in the tree is considered *pure* if 100% of cases in the node fall into a specific category of the target field. Target and input fields can be numeric ranges or categorical (nominal, ordinal, or flags); all splits are binary (only two subgroups).

Chapter 10. Statistical Models

Statistical models use mathematical equations to encode information extracted from the data. In some cases, statistical modeling techniques can provide adequate models very quickly. Even for problems in which more flexible machine-learning techniques (such as neural networks) can ultimately give better results, you can use some statistical models as baseline predictive models to judge the performance of more advanced techniques.

The following statistical modeling nodes are available.

Linear regression models predict a continuous target based on linear relationships between the target and one or more predictors.

Logistic regression is a statistical technique for classifying records based on values of input fields. It is analogous to linear regression but takes a categorical target field instead of a numeric range.

The PCA/Factor node provides powerful data-reduction techniques to reduce the complexity of your data. Principal components analysis (PCA) finds linear combinations of the input fields that do the best job of capturing the variance in the entire set of fields, where the components are orthogonal (perpendicular) to each other. Factor analysis attempts to identify underlying factors that explain the pattern of correlations within a set of observed fields. For both approaches, the goal is to find a small number of derived fields that effectively summarizes the information in the original set of fields.

Discriminant analysis makes more stringent assumptions than logistic regression but can be a valuable alternative or supplement to a logistic regression analysis when those assumptions are met.

The Generalized Linear model expands the general linear model so that the dependent variable is linearly related to the factors and covariates through a specified link function. Moreover, the model allows for the dependent variable to have a non-normal distribution. It covers the functionality of a wide number of statistical models, including linear regression, logistic regression, loglinear models for count data, and interval-censored survival models.

A generalized linear mixed model (GLMM) extends the linear model so that the target can have a non-normal distribution, is linearly related to the factors and covariates via a specified link function, and so that the observations can be correlated. Generalized linear mixed models cover a wide variety of models, from simple linear regression to complex multilevel models for non-normal longitudinal data.

The Cox regression node enables you to build a survival model for time-to-event data in the presence of censored records. The model produces a survival function that predicts the probability that the event of interest has occurred at a given time (*t*) for given values of the input variables.

Model selection

Model selection method. Choose one of the model selection methods (details below) or **Include all predictors**, which simply enters all available predictors as main effects model terms. By default, **Forward stepwise** is used.

Forward Stepwise Selection. This starts with no effects in the model and adds and removes effects one step at a time until no more can be added or removed according to the stepwise criteria.

- **Criteria for entry/removal.** This is the statistic used to determine whether an effect should be added to or removed from the model. **Information Criterion (AICC)** is based on the likelihood of the training set given the model, and is adjusted to penalize overly complex models. **F Statistics** is based on a statistical test of the improvement in model error. **Adjusted R-squared** is based on the fit of the training set, and is adjusted to penalize overly complex models. **Overfit Prevention Criterion (ASE)** is based on the fit (average squared error, or ASE) of the overfit prevention set. The overfit prevention set is a random subsample of approximately 30% of the original dataset that is not used to train the model.

  If any criterion other than **F Statistics** is chosen, then at each step the effect that corresponds to the greatest positive increase in the criterion is added to the model. Any effects in the model that correspond to a decrease in the criterion are removed.

  If **F Statistics** is chosen as the criterion, then at each step the effect that has the smallest *p*-value less than the specified threshold, **Include effects with p-values less than**, is added to the model. The default is 0.05. Any effects in the model with a *p*-value greater than the specified threshold, **Remove effects with p-values greater than**, are removed. The default is 0.10.
- **Customize maximum number of effects in the final model.** By default, all available effects can be entered into the model. Alternatively, if the stepwise algorithm ends a step with the specified maximum number of effects, the algorithm stops with the current set of effects.
- **Customize maximum number of steps.** The stepwise algorithm stops after a certain number of steps. By default, this is 3 times the number of available effects. Alternatively, specify a positive integer maximum number of steps.

Best Subsets Selection. This checks "all possible" models, or at least a larger subset of the possible models than forward stepwise, to choose the best according to the best subsets criterion. **Information Criterion (AICC)** is based on the likelihood of the training set given the model, and is adjusted to penalize overly complex models. **Adjusted R-squared** is based on the fit of the training set, and is adjusted to penalize overly complex models. **Overfit Prevention Criterion (ASE)** is based on the fit (average squared error, or ASE) of the overfit prevention set. The overfit prevention set is a random subsample of approximately 30% of the original dataset that is not used to train the model.

The model with the greatest value of the criterion is chosen as the best model.

Note: Best subsets selection is more computationally intensive than forward stepwise selection. When best subsets is performed in conjunction with boosting, bagging, or very large datasets, it can take considerably longer to build than a standard model built using forward stepwise selection.

Ensembles

These settings determine the behavior of ensembling that occurs when boosting, bagging, or very large datasets are requested in Objectives. Options that do not apply to the selected objective are ignored.

Bagging and Very Large Datasets. When scoring an ensemble, this is the rule used to combine the predicted values from the base models to compute the ensemble score value.

- **Default combining rule for continuous targets.** Ensemble predicted values for continuous targets can be combined using the mean or median of the predicted values from the base models.

Note that when the objective is to enhance model accuracy, the combining rule selections are ignored. Boosting always uses a weighted majority vote to score categorical targets and a weighted median to score continuous targets.

- **Merge categories to maximize association with target** identifies "similar" predictor categories based upon the relationship between the input and the target. Categories that are not significantly different (that is, having a *p*-value greater than 0.05) are merged.
- **Exclude constant predictor / after outlier handling / after merging of categories** removes predictors that have a single value, possibly after other ADP actions have been taken.

Predictor Importance

Typically, you will want to focus your modeling efforts on the predictor fields that matter most and consider dropping or ignoring those that matter least. The predictor importance chart helps you do this by indicating the relative importance of each predictor in estimating the model. Since the values are relative, the sum of the values for all predictors on the display is 1.0. Predictor importance does not relate to model accuracy. It just relates to the importance of each predictor in making a prediction, not whether or not the prediction is accurate.

Predicted By Observed

This displays a binned scatterplot of the predicted values on the vertical axis by the observed values on the horizontal axis. Ideally, the points should lie on a 45-degree line; this view can tell you whether any records are predicted particularly badly by the model.

Residuals

This displays a diagnostic chart of model residuals.

Chart styles. There are different display styles, which are accessible from the **Style** drop-down.

- **Histogram.** This is a binned histogram of the studentized residuals with an overlay of the normal distribution. Linear models assume that the residuals have a normal distribution, so the histogram should ideally closely approximate the smooth line.
- **P-P Plot.** This is a binned probability-probability plot comparing the studentized residuals to a normal distribution. If the slope of the plotted points is less steep than the normal line, the residuals show greater variability than a normal distribution; if the slope is steeper, the residuals show less variability than a normal distribution. If the plotted points have an S-shaped curve, then the distribution of residuals is skewed.

Outliers

This table lists records that exert undue influence upon the model, and displays the record ID (if specified on the Fields tab), target value, and Cook's distance. Cook's distance is a measure of how much the residuals of all records would change if a particular record were excluded from the calculation of the model coefficients. A large Cook's distance indicates that excluding a record from changes the coefficients substantially, and should therefore be considered influential.

Influential records should be examined carefully to determine whether you can give them less weight in estimating the model, or truncate the outlying values to some acceptable threshold, or remove the influential records completely.

Effects

This view displays the size of each effect in the model.

Styles. There are different display styles, which are accessible from the **Style** dropdown list.

- **Diagram.** This is a chart in which effects are sorted from top to bottom by decreasing predictor importance. Connecting lines in the diagram are weighted based on effect significance, with greater line width corresponding to more significant effects (smaller *p*-values). Hovering over a connecting line reveals a tooltip that shows the *p*-value and importance of the effect. This is the default.
- **Table.** This is an ANOVA table for the overall model and the individual model effects. The individual effects are sorted from top to bottom by decreasing predictor importance. Note that by default, the

table is collapsed to only show the results for the overall model. To see the results for the individual model effects, click the **Corrected Model** cell in the table.

Predictor importance. There is a Predictor Importance slider that controls which predictors are shown in the view. This does not change the model, but simply allows you to focus on the most important predictors. By default, the top 10 effects are displayed.

Significance. There is a Significance slider that further controls which effects are shown in the view, beyond those shown based on predictor importance. Effects with significance values greater than the slider value are hidden. This does not change the model, but simply allows you to focus on the most important effects. By default the value is 1.00, so that no effects are filtered based on significance.

Coefficients

This view displays the value of each coefficient in the model. Note that factors (categorical predictors) are indicator-coded within the model, so that **effects** containing factors will generally have multiple associated **coefficients**; one for each category except the category corresponding to the redundant (reference) parameter.

Styles. There are different display styles, which are accessible from the **Style** dropdown list.

- **Diagram.** This is a chart which displays the intercept first, and then sorts effects from top to bottom by decreasing predictor importance. Within effects containing factors, coefficients are sorted by ascending order of data values. Connecting lines in the diagram are colored based on the sign of the coefficient (see the diagram key) and weighted based on coefficient significance, with greater line width corresponding to more significant coefficients (smaller *p*-values). Hovering over a connecting line reveals a tooltip that shows the value of the coefficient, its *p*-value, and the importance of the effect the parameter is associated with. This is the default style.
- **Table.** This shows the values, significance tests, and confidence intervals for the individual model coefficients. After the intercept, the effects are sorted from top to bottom by decreasing predictor importance. Within effects containing factors, coefficients are sorted by ascending order of data values. Note that by default the table is collapsed to only show the coefficient, significance, and importance of each model parameter. To see the standard error, *t* statistic, and confidence interval, click the **Coefficient** cell in the table. Hovering over the name of a model parameter in the table reveals a tooltip that shows the name of the parameter, the effect the parameter is associated with, and (for categorical predictors), the value labels associated with the model parameter. This can be particularly useful to see the new categories created when automatic data preparation merges similar categories of a categorical predictor.

Predictor importance. There is a Predictor Importance slider that controls which predictors are shown in the view. This does not change the model, but simply allows you to focus on the most important predictors. By default, the top 10 effects are displayed.

Significance. There is a Significance slider that further controls which coefficients are shown in the view, beyond those shown based on predictor importance. Coefficients with significance values greater than the slider value are hidden. This does not change the model, but simply allows you to focus on the most important coefficients. By default the value is 1.00, so that no coefficients are filtered based on significance.

Estimated means

These are charts displayed for significant predictors. The chart displays the model-estimated value of the target on the vertical axis for each value of the predictor on the horizontal axis, holding all other predictors constant. It provides a useful visualization of the effects of each predictor's coefficients on the target.

Note: If no predictors are significant, no estimated means are produced.

Model Building Summary

When a model selection algorithm other than **None** is chosen on the Model Selection settings, this provides some details of the model building process.